



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH
DEPARTMENT OF COMPUTER SCIENCE
FALL 22-23

PROJECT NAME : BUS MANAGEMENT SYSTEM

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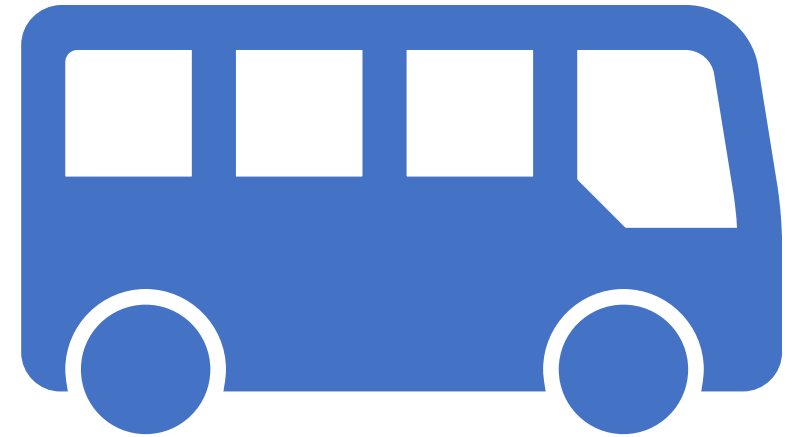
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INTRODUCTION :

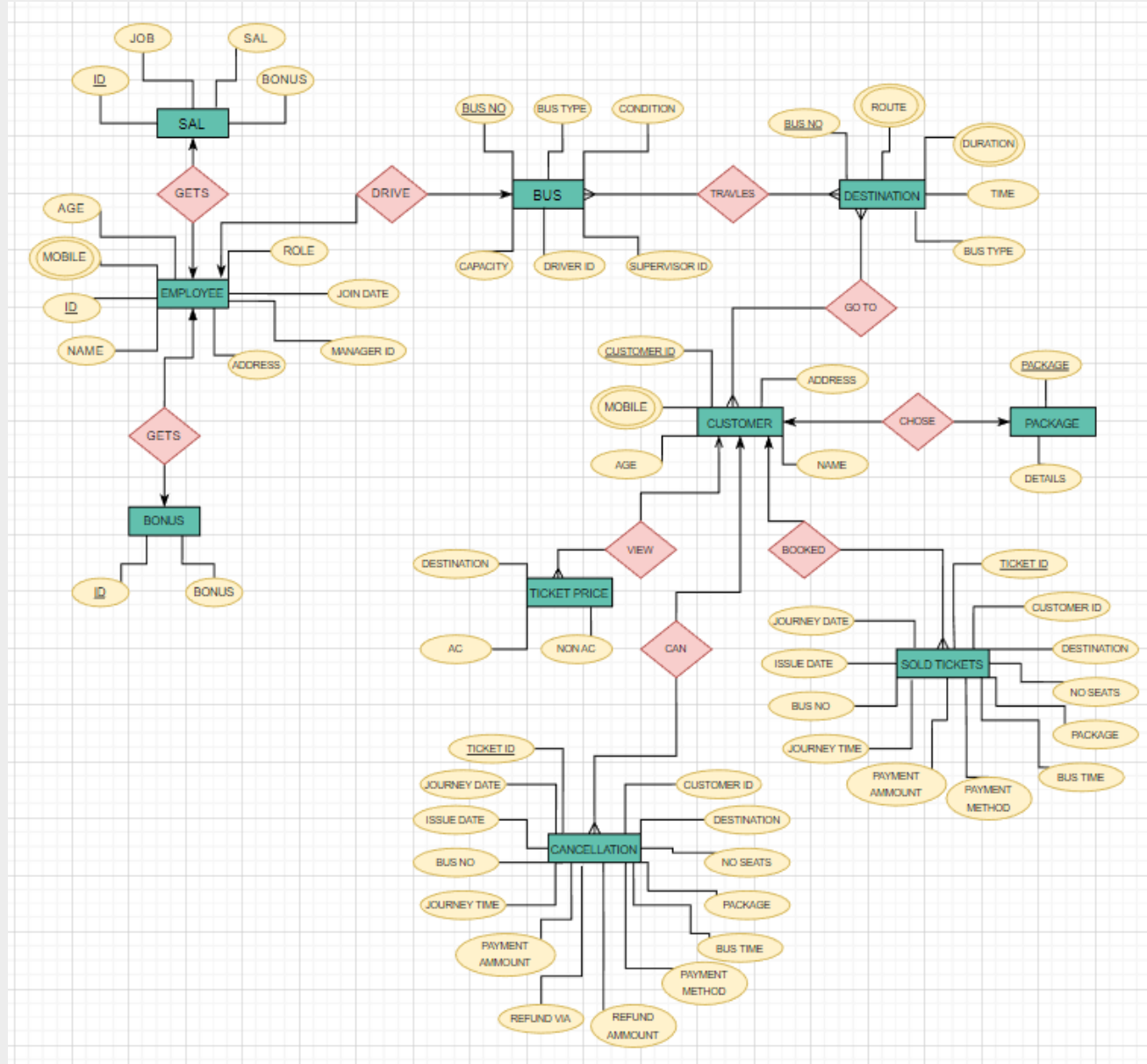
This project aims to create a suitable database system to manage and track the operations of a bus transportation company. The database will store information on routes, schedules, buses, drivers, and passengers, and will help to analyze and optimize the company's operations. It allows easy access to the information and will help to the processes of scheduling, dispatch, and maintenance. We hope that this project will greatly improve the efficiency and effectiveness of the bus transportation company, and provide a easy management and planning of its operations.



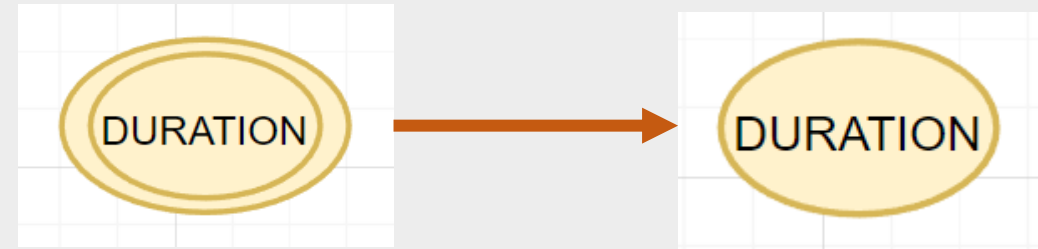
SECENARIO :

In a bus company, there are many employees. Every employee has their unique id, name, age, address and phone number, role, join date and manager id. Every employee gets salary and bonus. Driver drives buses. Every bus has a unique bus number, bus type, capacity, driver and supervisor id. Bus and Customers goes to destination and every destination has a route, duration and so on. Every customer has a unique id, name, address, age, mobile number. Customer can view ticket prices according to the destination. Customers can buy tickets and Packages. Every sold tickets has a unique ticket id, customer id, bus time, journey date, bus no, payment amount and payment method and so on. A customer can also cancel a ticket. A canceled ticket holder will get refund but 10% amount will be deducted.

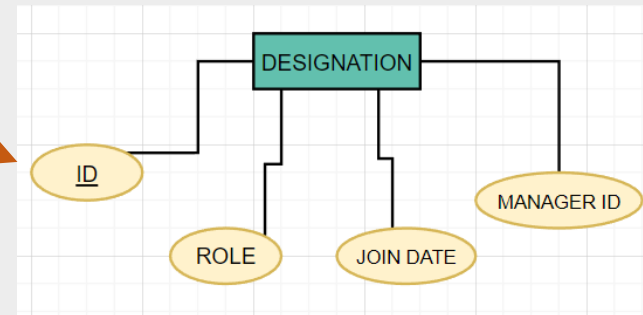
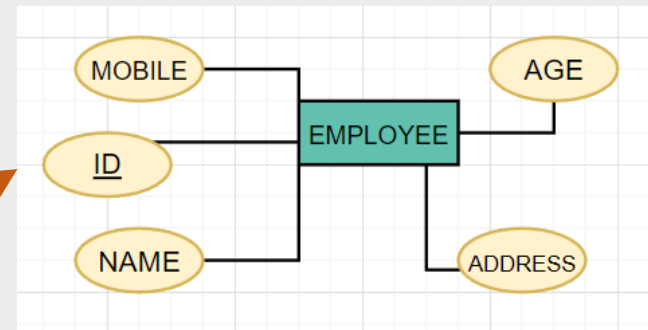
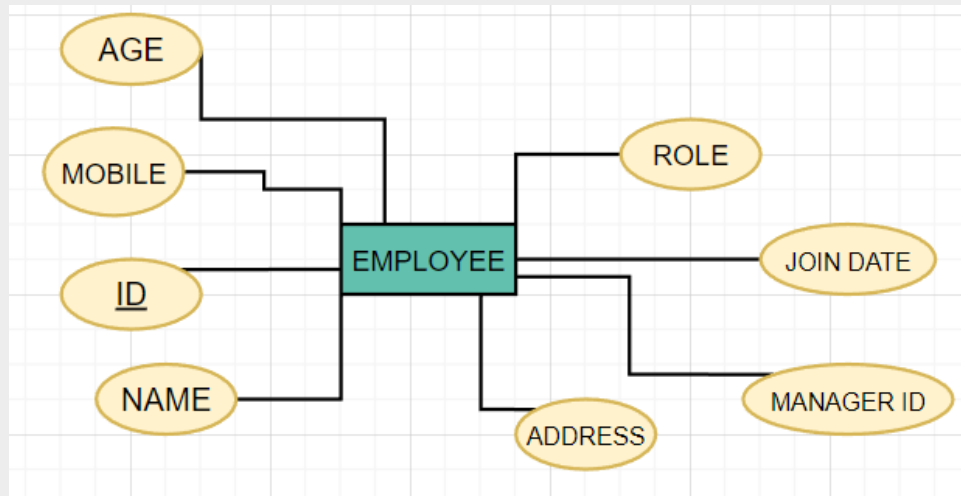
ER DIAGRAM : BEFORE NORMALIZATION



NORMALIZING

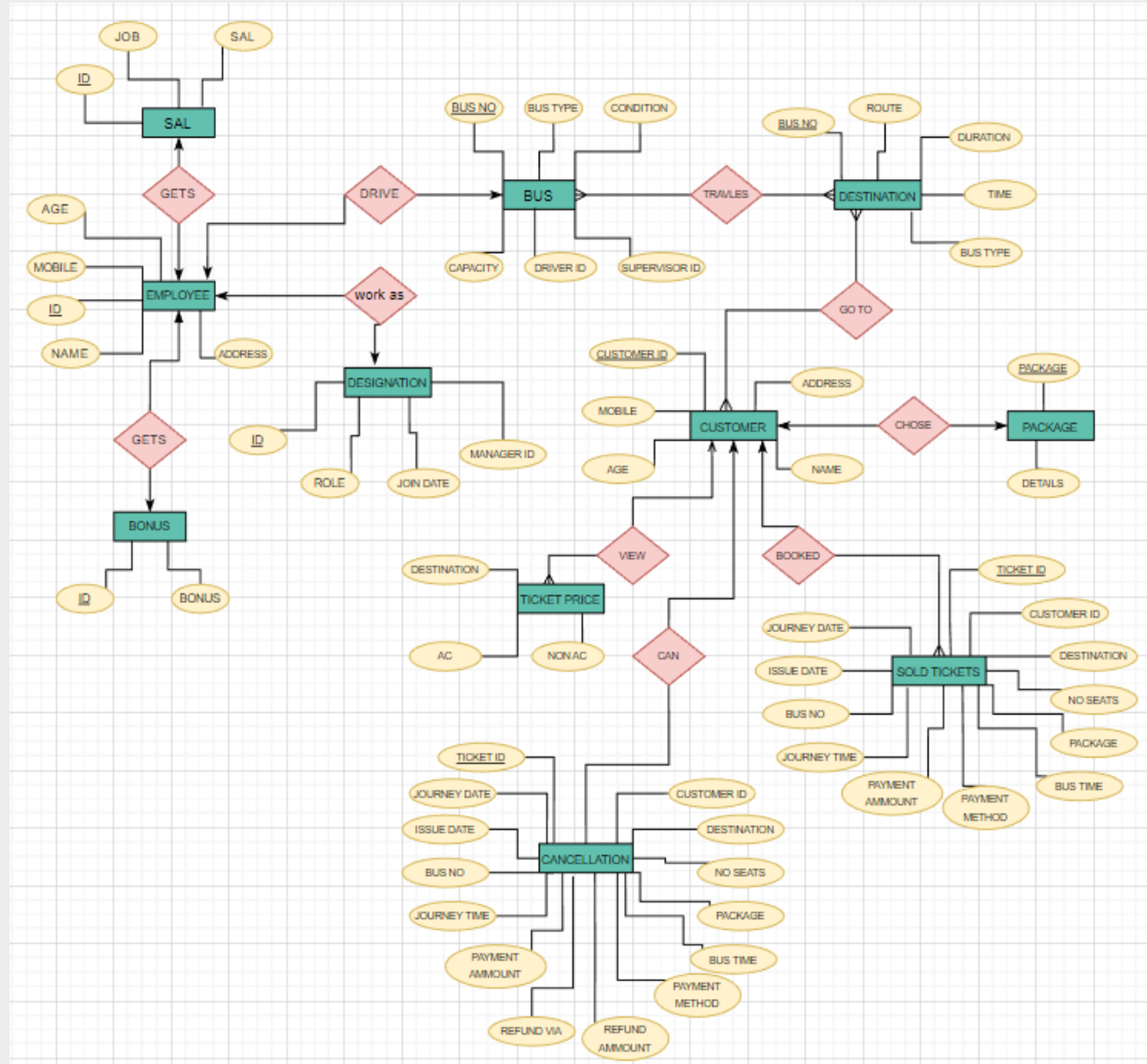


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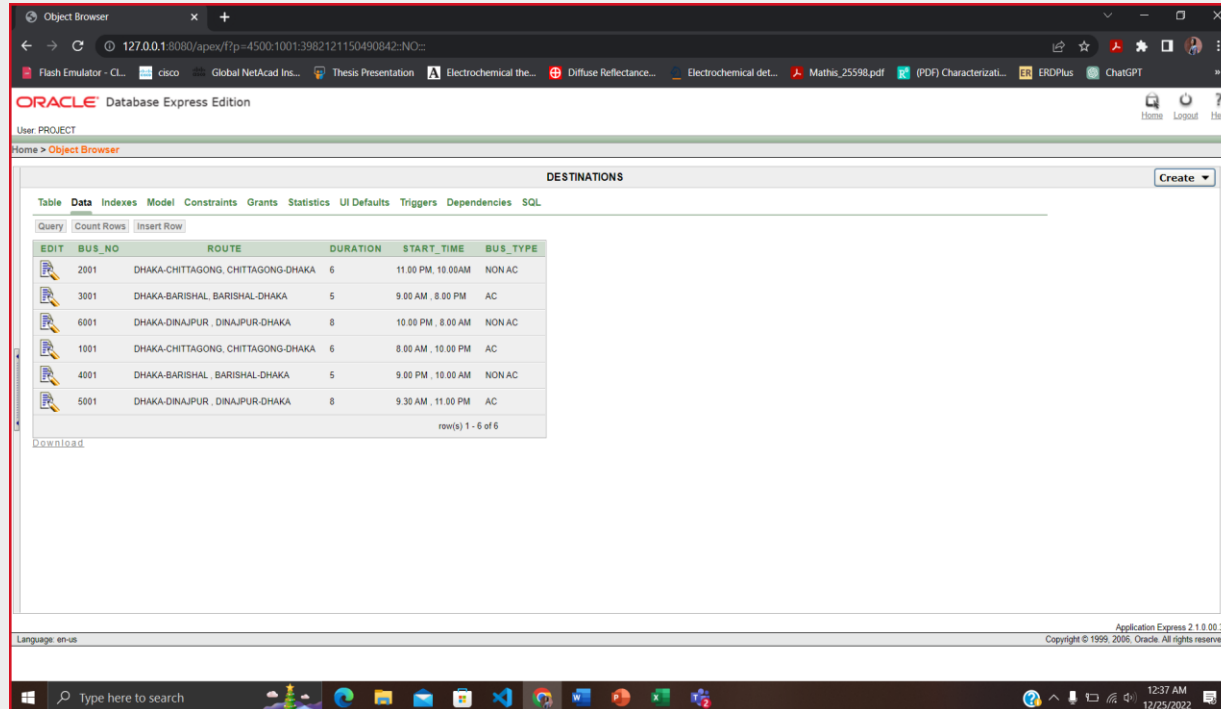
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ER DIAGRAM : AFTER NORMALIZATION



NORMALIZATION TABLE

1NF



ORACLE Database Express Edition

User PROJECT

Home > Object Browser

DESTINATIONS

Create

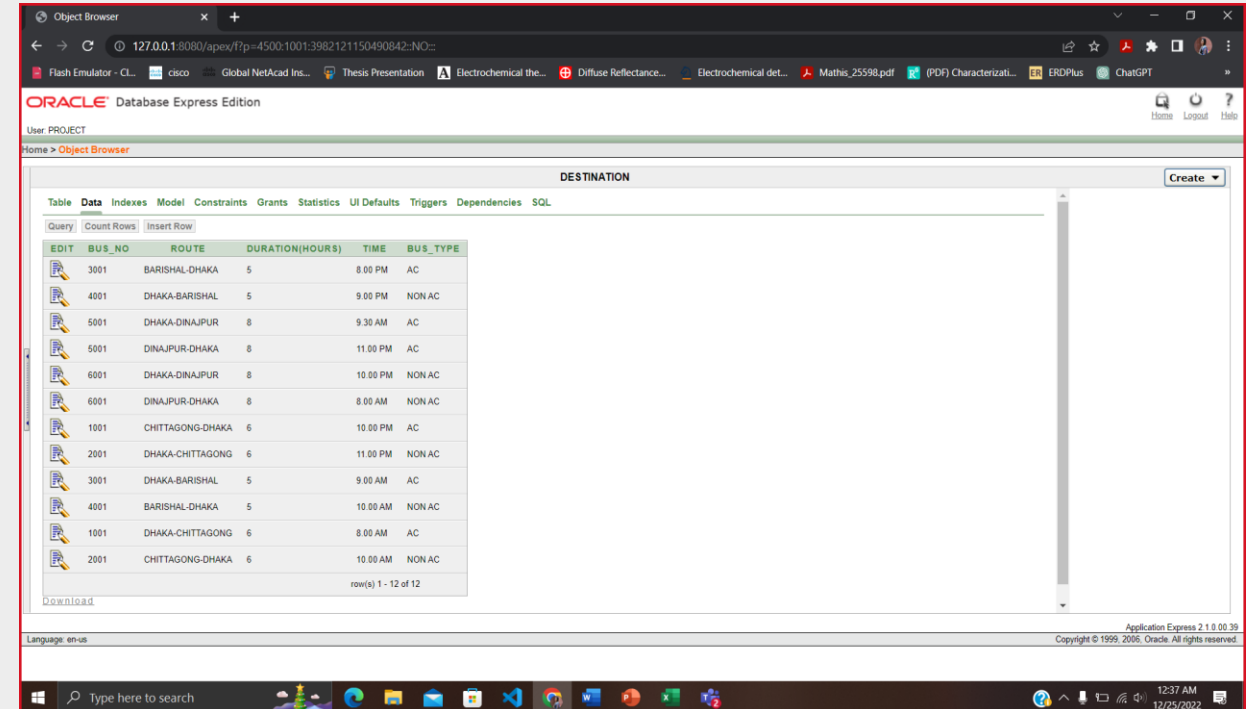
EDIT	BUS_NO	ROUTE	DURATION	START_TIME	BUS_TYPE
	2001	DHAKA-CHITTAGONG, CHITTAGONG-DHAKA	6	11:00 PM, 10:00 AM	NON AC
	3001	DHAKA-BARISHAL, BARISHAL-DHAKA	5	9:00 AM, 8:00 PM	AC
	6001	DHAKA-DINAJPUR, DINAJPUR-DHAKA	8	10:00 PM, 8:00 AM	NON AC
	1001	DHAKA-CHITTAGONG, CHITTAGONG-DHAKA	6	8:00 AM, 10:00 PM	AC
	4001	DHAKA-BARISHAL, BARISHAL-DHAKA	5	9:00 PM, 10:00 AM	NON AC
	5001	DHAKA-DINAJPUR, DINAJPUR-DHAKA	8	9:30 AM, 11:00 PM	AC

row(s) 1 - 6 of 6

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BEFORE NORMALIZATION



ORACLE Database Express Edition

User PROJECT

Home > Object Browser

DESTINATION

Create

EDIT	BUS_NO	ROUTE	DURATION(HOURS)	TIME	BUS_TYPE
	3001	BARISHAL-DHAKA	5	8:00 PM	AC
	4001	DHAKA-BARISHAL	5	9:00 PM	NON AC
	5001	DHAKA-DINAJPUR	8	9:30 AM	AC
	5001	DINAJPUR-DHAKA	8	11:00 PM	AC
	6001	DHAKA-DINAJPUR	8	10:00 PM	NON AC
	6001	DINAJPUR-DHAKA	8	8:00 AM	NON AC
	1001	CHITTAGONG-DHAKA	6	10:00 PM	AC
	2001	DHAKA-CHITTAGONG	6	11:00 PM	NON AC
	3001	DHAKA-BARISHAL	5	9:00 AM	AC
	4001	BARISHAL-DHAKA	5	10:00 AM	NON AC
	1001	DHAKA-CHITTAGONG	6	8:00 AM	AC
	2001	CHITTAGONG-DHAKA	6	10:00 AM	NON AC

row(s) 1 - 12 of 12

Download

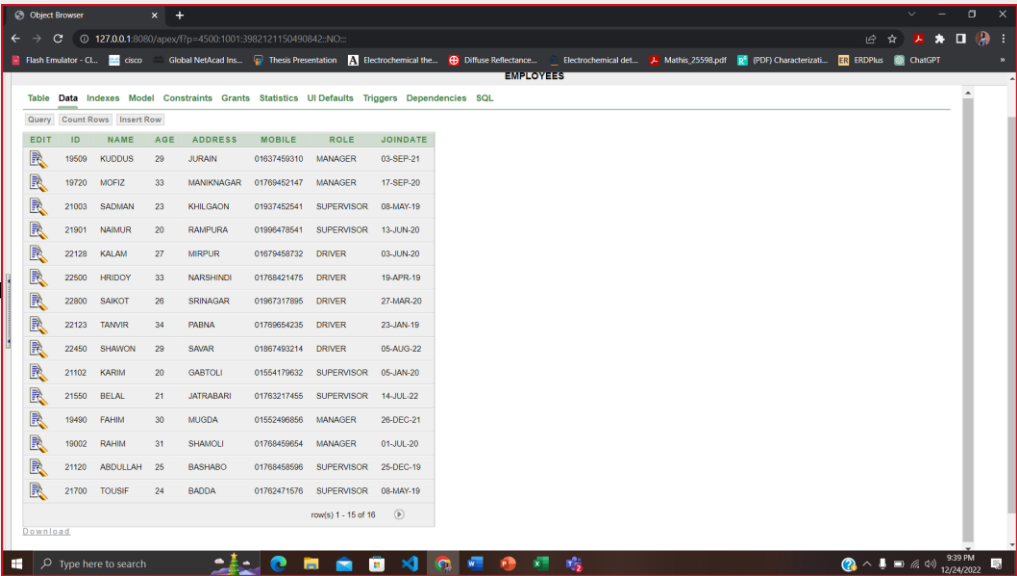
Application Express 2.1.0.00.39
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AFTER NORMALIZATION

NORMALIZATION TABLE

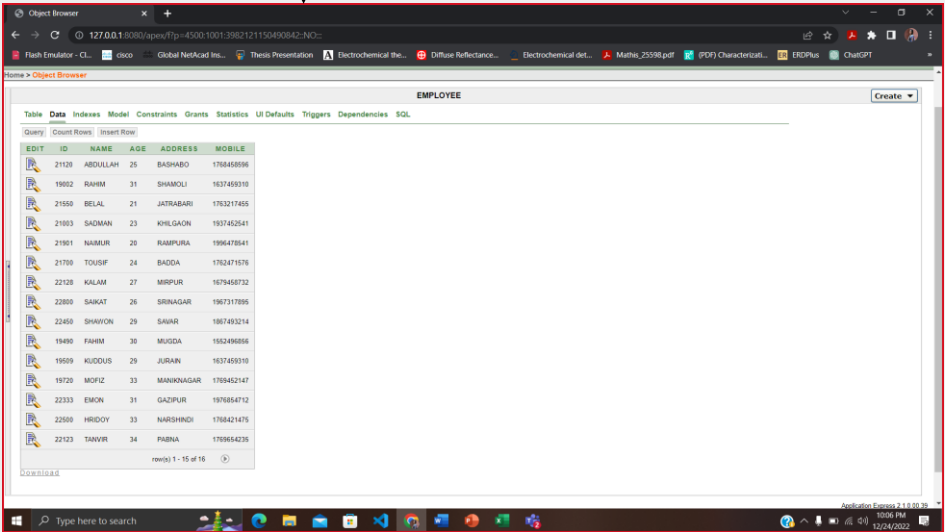
2NF

BEFORE NORMALIZATION

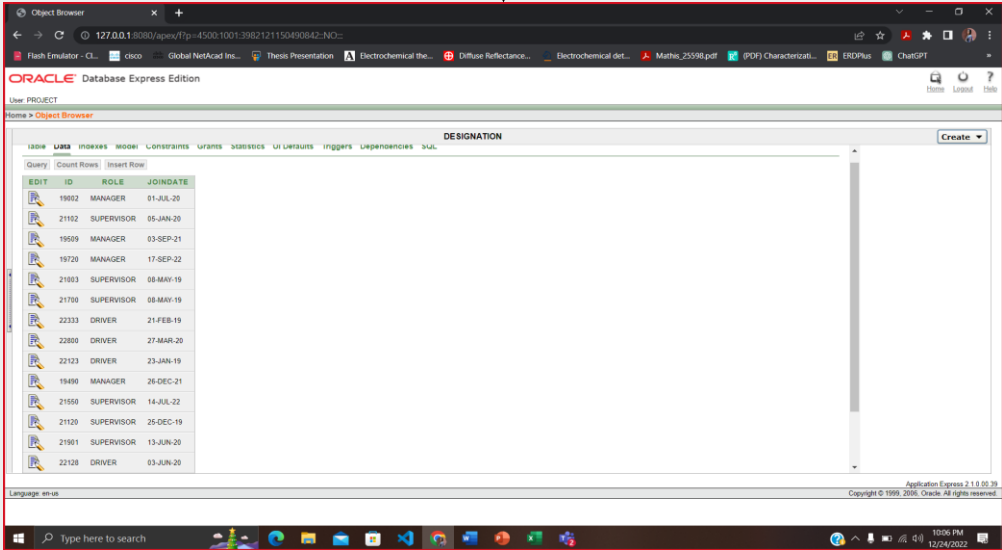


EDIT	ID	NAME	AGE	ADDRESS	MOBILE	ROLE	JOINDATE
	19509	KUDOUS	29	JURAIN	01637459310	MANAGER	03-SEP-21
	19720	MOFIZ	33	MANIKNAGAR	01769452147	MANAGER	17-SEP-20
	21003	SADMAN	23	KHILGAON	01837452541	SUPERVISOR	08-MAY-19
	21901	NAMJUR	20	RAMPURA	01996478541	SUPERVISOR	13-JUN-20
	22128	KALAM	27	MIRPUR	01679458732	DRIVER	03-JUN-20
	22500	HRIDOY	33	NARSHINDI	01768421475	DRIVER	19-APR-19
	22800	SAKOT	26	SRINAGAR	01967317895	DRIVER	27-MAR-20
	22123	TANVIR	34	PABNA	01769654235	DRIVER	23-JAN-19
	22450	SHAWON	29	SAVAR	01867493214	DRIVER	05-AUG-22
	21102	KARIM	20	GABTOLI	01554179632	SUPERVISOR	05-JAN-20
	21550	BELAL	21	JATRABARI	01763217455	SUPERVISOR	14-JUL-22
	19490	FAHM	30	MUGDA	01552496956	MANAGER	26-DEC-21
	19002	RAHM	31	SHAMOLI	01768459954	MANAGER	01-JUL-20
	21120	ABOULLAH	25	BASHABO	01768458996	SUPERVISOR	25-DEC-19
	21700	TOUSIF	24	BADDA	01762471576	SUPERVISOR	08-MAY-19

AFTER
NORMALIZATION



EDIT	ID	NAME	AGE	ADDRESS	MOBILE
	21120	ABOULLAH	25	BASHABO	01768458996
	19002	RAHM	31	SHAMOLI	01768459954
	21550	BELAL	21	JATRABARI	01763217455
	21003	SADMAN	23	KHILGAON	01837452541
	21901	NAMJUR	20	RAMPURA	01996478541
	21700	TOUSIF	24	BADDA	01762471576
	22128	KALAM	27	MIRPUR	01679458732
	22800	SAKOT	26	SRINAGAR	01967317895
	22450	SHAWON	29	SAVAR	01867493214
	19490	FAHM	30	MUGDA	01552496956
	19509	KUDOUS	29	JURAIN	01637459310
	19720	MOFIZ	33	MANIKNAGAR	01769452147
	22333	EMON	31	GAZIPUR	01768547412
	22058	HRIDOY	33	NARSHINDI	01768421475
	22123	TANVIR	34	PABNA	01769654235

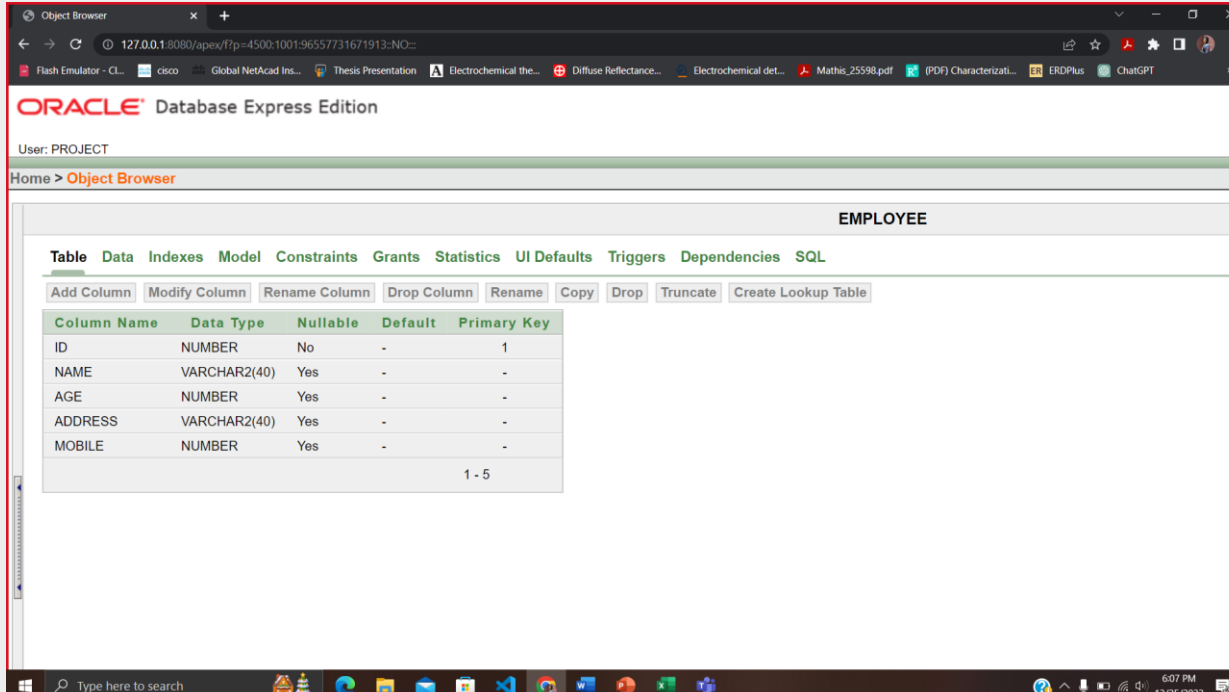


EDIT	ID	ROLE	JOINDATE
	19902	MANAGER	01-JUL-20
	21102	SUPERVISOR	05-JAN-20
	19509	MANAGER	03-SEP-21
	19720	MANAGER	17-SEP-22
	21003	SUPERVISOR	08-MAY-19
	21700	SUPERVISOR	08-MAY-19
	22333	DRIVER	21-FEB-19
	22800	DRIVER	27-MAR-20
	22123	DRIVER	23-JAN-19
	19490	MANAGER	26-DEC-21
	21550	SUPERVISOR	14-JUL-22
	21120	SUPERVISOR	25-DEC-19
	21901	SUPERVISOR	13-JUN-20
	22128	DRIVER	03-JUN-20

Table creation

EMPLOYEE

```
CREATE TABLE "EMPLOYEE"  
(  
  "ID" NUMBER NOT NULL ENABLE,  
  "NAME" VARCHAR2(40),  
  "AGE" NUMBER,  
  "ADDRESS" VARCHAR2(40),  
  "MOBILE" NUMBER,  
  CONSTRAINT "EMPLOYEE_PK" PRIMARY KEY ("ID")  
ENABLE  
)
```



The screenshot shows the Oracle Database Express Edition Object Browser interface. The 'EMPLOYEE' table is selected, and the 'Table' tab is active. The table structure is displayed in a table with the following columns: Column Name, Data Type, Nullable, Default, and Primary Key.

Column Name	Data Type	Nullable	Default	Primary Key
ID	NUMBER	No	-	1
NAME	VARCHAR2(40)	Yes	-	-
AGE	NUMBER	Yes	-	-
ADDRESS	VARCHAR2(40)	Yes	-	-
MOBILE	NUMBER	Yes	-	-

1 - 5

Table creation

BUS

```
CREATE TABLE "BUS"
```

```
(
```

```
    "BUS_NO" NUMBER,
```

```
    "BUS_TYPE" VARCHAR2(40),
```

```
    "CAPACITY" NUMBER,
```

```
    "CONDITION" VARCHAR2(40),
```

```
    "DRIVER_ID" NUMBER,
```

```
    "SUPERVISOR_ID" NUMBER,
```

```
    CONSTRAINT "BUS_PK" PRIMARY KEY ("BUS_NO") ENABLE,
```

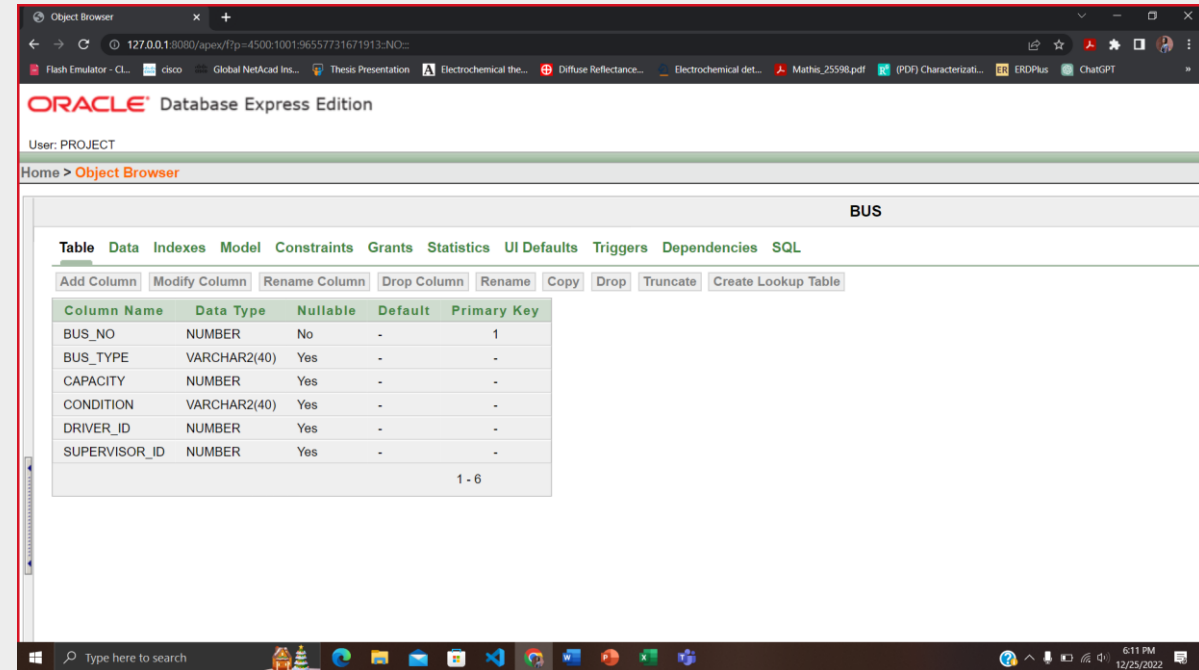
```
    CONSTRAINT "BUS_FK" FOREIGN KEY ("DRIVER_ID")
```

```
        REFERENCES "EMPLOYEE" ("ID") ENABLE,
```

```
    CONSTRAINT "BUS_FK2" FOREIGN KEY ("SUPERVISOR_ID")
```

```
        REFERENCES "EMPLOYEE" ("ID") ENABLE
```

```
)
```



The screenshot shows the Oracle Database Express Edition Object Browser interface. The table 'BUS' is selected, and its structure is displayed in a table format. The table has six columns: BUS_NO, BUS_TYPE, CAPACITY, CONDITION, DRIVER_ID, and SUPERVISOR_ID. BUS_NO is the primary key. The table is located in the schema 'PROJECT'.

Column Name	Data Type	Nullable	Default	Primary Key
BUS_NO	NUMBER	No	-	1
BUS_TYPE	VARCHAR2(40)	Yes	-	-
CAPACITY	NUMBER	Yes	-	-
CONDITION	VARCHAR2(40)	Yes	-	-
DRIVER_ID	NUMBER	Yes	-	-
SUPERVISOR_ID	NUMBER	Yes	-	-

Table creation

PACKAGE

```
CREATE TABLE "PACKAGE"  
(  
    "PACKAGE" VARCHAR2(40),  
    "DETAILS" VARCHAR2(40),  
    CONSTRAINT "PACKAGE_PK" PRIMARY KEY ("PACKAGE")  
ENABLE  
)
```

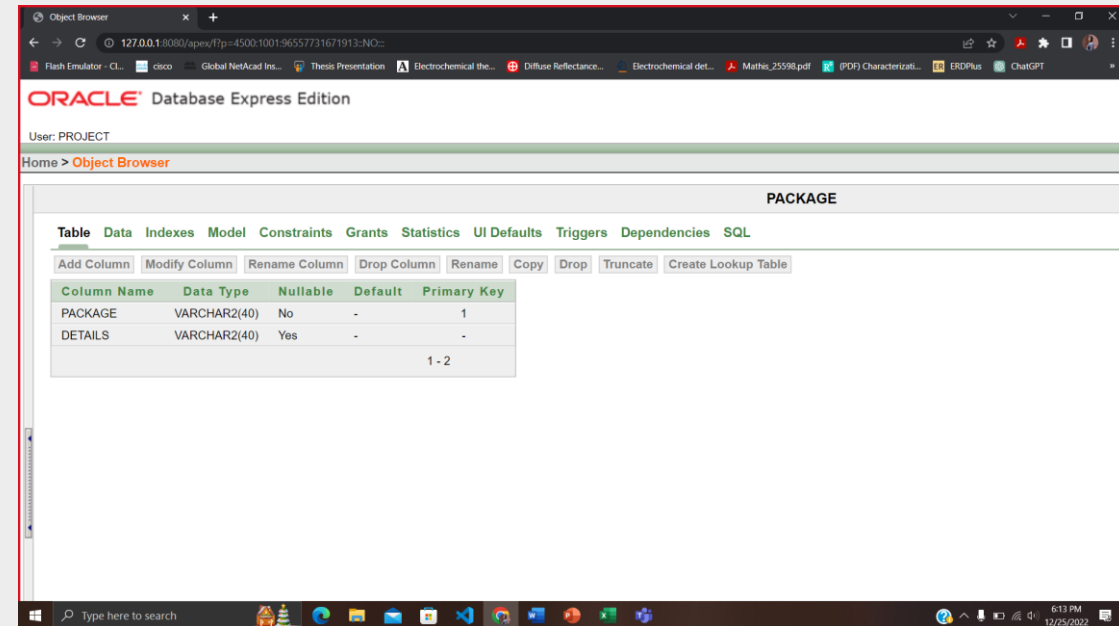
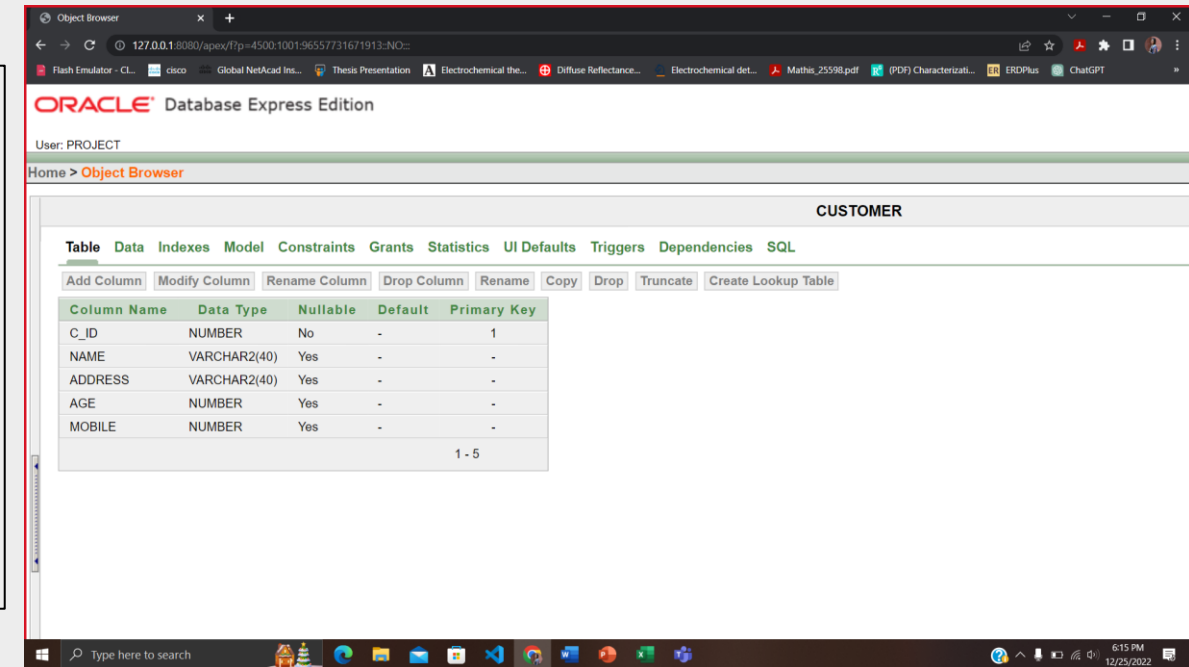


Table creation

CUSTOMER

```
CREATE TABLE "CUSTOMER"  
(  
    "C_ID" NUMBER NOT NULL ENABLE,  
    "NAME" VARCHAR2(40),  
    "ADDRESS" VARCHAR2(40),  
    "AGE" NUMBER,  
    "MOBILE" NUMBER,  
    CONSTRAINT "CUSTOMER_PK" PRIMARY KEY ("C_ID")  
ENABLE  
)
```



The screenshot shows the Oracle Database Express Edition Object Browser interface. The user is logged in as 'PROJECT'. The 'Object Browser' tab is selected, showing the 'CUSTOMER' table. The table structure is displayed in a table format with columns: Column Name, Data Type, Nullable, Default, and Primary Key. The table has five columns: C_ID (NUMBER, Not Null, Primary Key), NAME (VARCHAR2(40), Yes), ADDRESS (VARCHAR2(40), Yes), AGE (NUMBER, Yes), and MOBILE (NUMBER, Yes). The table is highlighted in the Object Browser tree.

Column Name	Data Type	Nullable	Default	Primary Key
C_ID	NUMBER	No	-	1
NAME	VARCHAR2(40)	Yes	-	-
ADDRESS	VARCHAR2(40)	Yes	-	-
AGE	NUMBER	Yes	-	-
MOBILE	NUMBER	Yes	-	-

Table creation

DESIGNATION

```
CREATE TABLE "DESIGNATION"  
(  
    "ID" NUMBER NOT NULL ENABLE,  
    "ROLE" VARCHAR2(40),  
    "JOINDATE" DATE,  
    "MANAGER_ID" NUMBER,  
    CONSTRAINT "DESIGNATION_PK" PRIMARY KEY ("ID") ENABLE,  
    CONSTRAINT "DESIGNATION_FK" FOREIGN KEY ("ID")  
        REFERENCES "EMPLOYEE" ("ID") ENABLE  
)
```

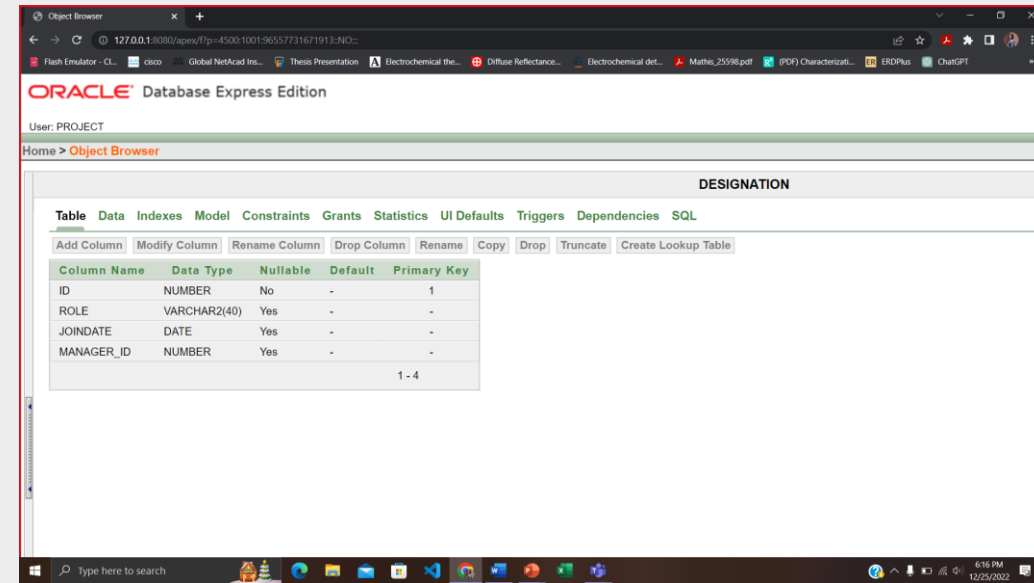
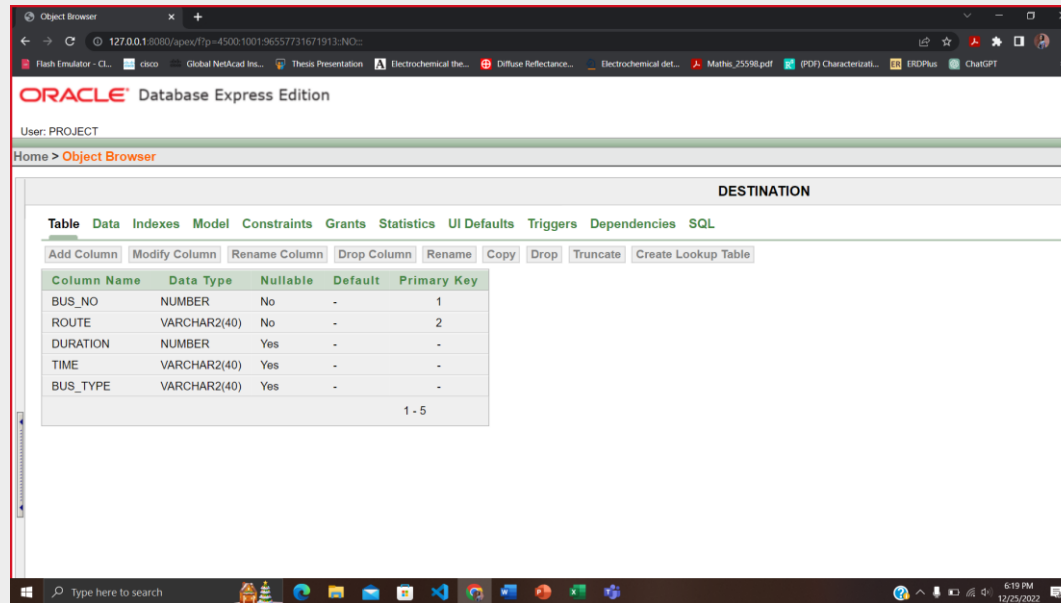


Table creation

DESTINATION

```
CREATE TABLE "DESTINATION"  
(  
    "BUS_NO" NUMBER,  
    "ROUTE" VARCHAR2(40),  
    "DURATION" NUMBER,  
    "TIME" VARCHAR2(40),  
    "BUS_TYPE" VARCHAR2(40),  
    CONSTRAINT "DESTINATION_PK" PRIMARY KEY ("BUS_NO", "ROUTE") ENABLE,  
    CONSTRAINT "DESTINATION_FK" FOREIGN KEY ("BUS_NO")  
    REFERENCES "BUS" ("BUS_NO") ENABLE  
)
```



The screenshot shows the Oracle Database Express Edition Object Browser interface. The table 'DESTINATION' is selected, and the 'Table' tab is active. The table structure is displayed in a table with columns: Column Name, Data Type, Nullable, Default, and Primary Key. The table has five columns: BUS_NO (NUMBER, Not Nullable, Default -), ROUTE (VARCHAR2(40), Not Nullable, Default -), DURATION (NUMBER, Nullable, Default -), TIME (VARCHAR2(40), Nullable, Default -), and BUS_TYPE (VARCHAR2(40), Nullable, Default -). The primary key is defined on BUS_NO and ROUTE.

Column Name	Data Type	Nullable	Default	Primary Key
BUS_NO	NUMBER	No	-	1
ROUTE	VARCHAR2(40)	No	-	2
DURATION	NUMBER	Yes	-	-
TIME	VARCHAR2(40)	Yes	-	-
BUS_TYPE	VARCHAR2(40)	Yes	-	-

Table creation

SAL

```
CREATE TABLE "SAL"  
(  
    "ID" NUMBER NOT NULL ENABLE,  
    "ROLE" VARCHAR2(40),  
    "SAL" VARCHAR2(40),  
    CONSTRAINT "SAL_PK" PRIMARY KEY ("ID") ENABLE,  
    CONSTRAINT "SAL_FK" FOREIGN KEY ("ID")  
        REFERENCES "EMPLOYEE" ("ID") ENABLE  
)
```

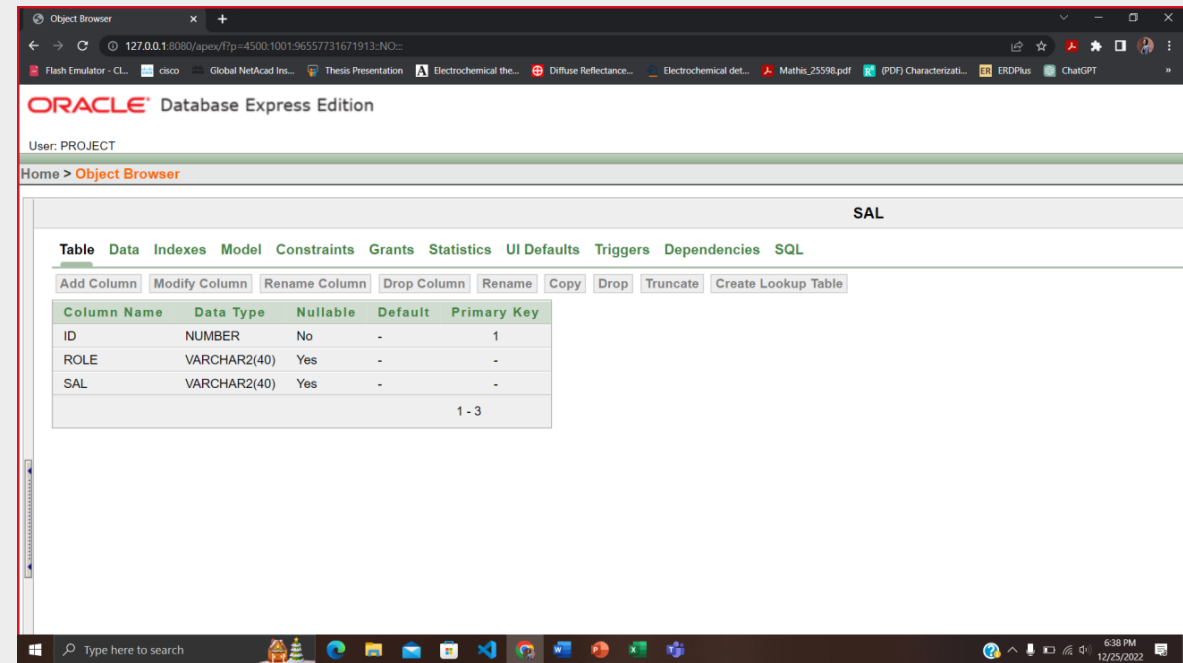


Table creation

BONUS

```
CREATE TABLE "BONUS"  
(  
    "ID" NUMBER,  
    "BONUS" VARCHAR2(40),  
    CONSTRAINT "BONUS_PK" PRIMARY KEY ("ID")  
    ENABLE,  
    CONSTRAINT "BONUS_FK" FOREIGN KEY ("ID")  
    REFERENCES "EMPLOYEE" ("ID") ENABLE  
)
```

The screenshot shows the Oracle Database Express Edition Object Browser interface. The browser is displaying the structure of the 'BONUS' table. The table has three columns: 'ID' (NUMBER), 'ROLE' (VARCHAR2(40)), and 'SAL' (VARCHAR2(40)). The 'ID' column is the primary key. The 'ROLE' and 'SAL' columns are nullable. The interface includes a navigation bar with tabs for Table, Data, Indexes, Model, Constraints, Grants, Statistics, UI Defaults, Triggers, Dependencies, and SQL. Below the navigation bar, there are buttons for Add Column, Modify Column, Rename Column, Drop Column, Rename, Copy, Drop, Truncate, and Create Lookup Table. The table structure is displayed in a table format with columns: Column Name, Data Type, Nullable, Default, and Primary Key.

Column Name	Data Type	Nullable	Default	Primary Key
ID	NUMBER	No	-	1
ROLE	VARCHAR2(40)	Yes	-	-
SAL	VARCHAR2(40)	Yes	-	-

1 - 3

Table creation

TICKET PRICE

```
CREATE TABLE "TICKET_PRICE"  
(  
    "DESTINATION" VARCHAR2(40),  
    "AC" NUMBER,  
    "NON_AC" NUMBER  
)
```

The screenshot displays the Oracle Database Express Edition Object Browser interface. The browser shows the 'TICKET_PRICE' table structure with the following columns:

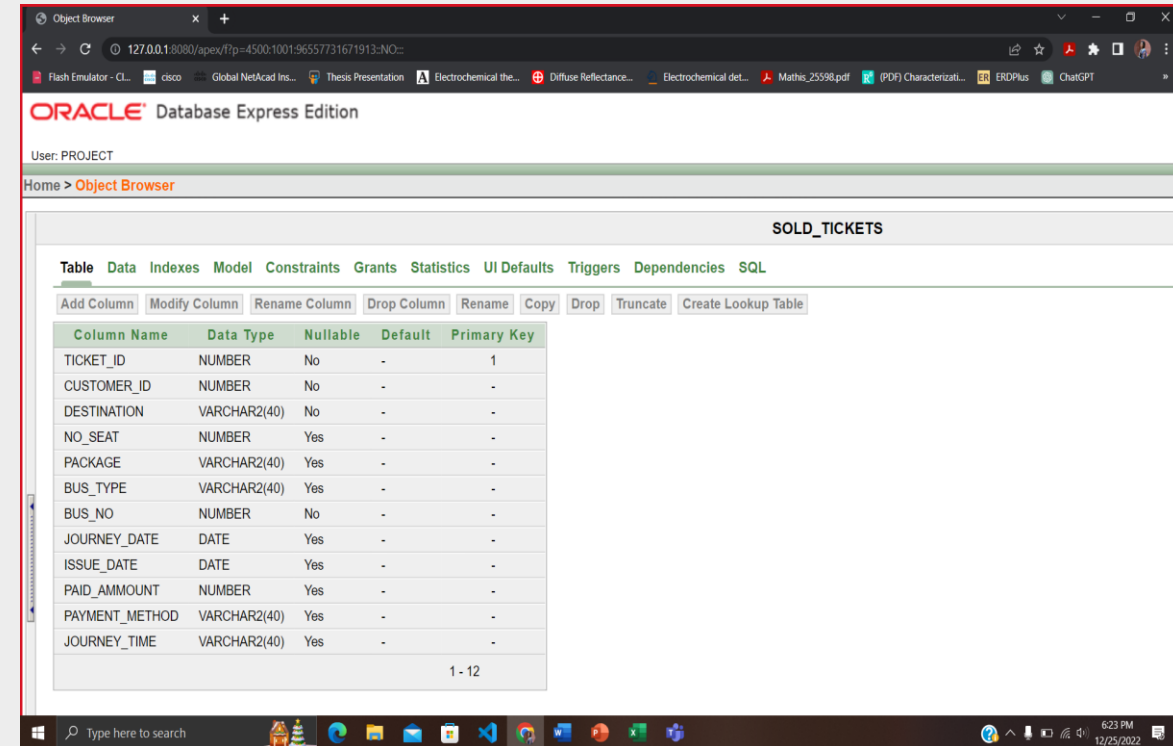
Column Name	Data Type	Nullable	Default	Primary Key
DESTINATION	VARCHAR2(40)	Yes	-	-
AC	NUMBER	Yes	-	-
NON_AC	NUMBER	Yes	-	-

The interface also includes navigation tabs for Table, Data, Indexes, Model, Constraints, Grants, Statistics, UI Defaults, Triggers, Dependencies, and SQL. Below the table structure, there are buttons for 'Add Column', 'Modify Column', 'Rename Column', 'Drop Column', 'Rename', 'Copy', 'Drop', 'Truncate', and 'Create Lookup Table'. The bottom of the window shows a Windows taskbar with various application icons and the system clock indicating 6:23 PM on 12/25/2022.

Table creation

SOLD TICKETS

```
CREATE TABLE "SOLD_TICKETS"
(
    "TICKET_ID" NUMBER NOT NULL ENABLE,
    "CUSTOMER_ID" NUMBER NOT NULL ENABLE,
    "DESTINATION" VARCHAR2(40) NOT NULL ENABLE,
    "NO_SEAT" NUMBER,
    "PACKAGE" VARCHAR2(40),
    "BUS_TYPE" VARCHAR2(40),
    "BUS_NO" NUMBER NOT NULL ENABLE,
    "JOURNEY_DATE" DATE,
    "ISSUE_DATE" DATE,
    "PAID_AMMOUNT" NUMBER,
    "PAYMENT_METHOD" VARCHAR2(40),
    "JOURNEY_TIME" VARCHAR2(40),
    CONSTRAINT "SOLD_TICKETS_PK" PRIMARY KEY ("TICKET_ID") ENABLE,
    CONSTRAINT "SOLD_TICKETS_FK" FOREIGN KEY ("CUSTOMER_ID")
    REFERENCES "CUSTOMER" ("C_ID") ENABLE,
    CONSTRAINT "SOLD_TICKETS_FK3" FOREIGN KEY ("PACKAGE")
    REFERENCES "PACKAGE" ("PACKAGE") ENABLE,
    CONSTRAINT "SOLD_TICKETS_FK4" FOREIGN KEY ("BUS_NO")
    REFERENCES "BUS" ("BUS_NO") ENABLE
)
```



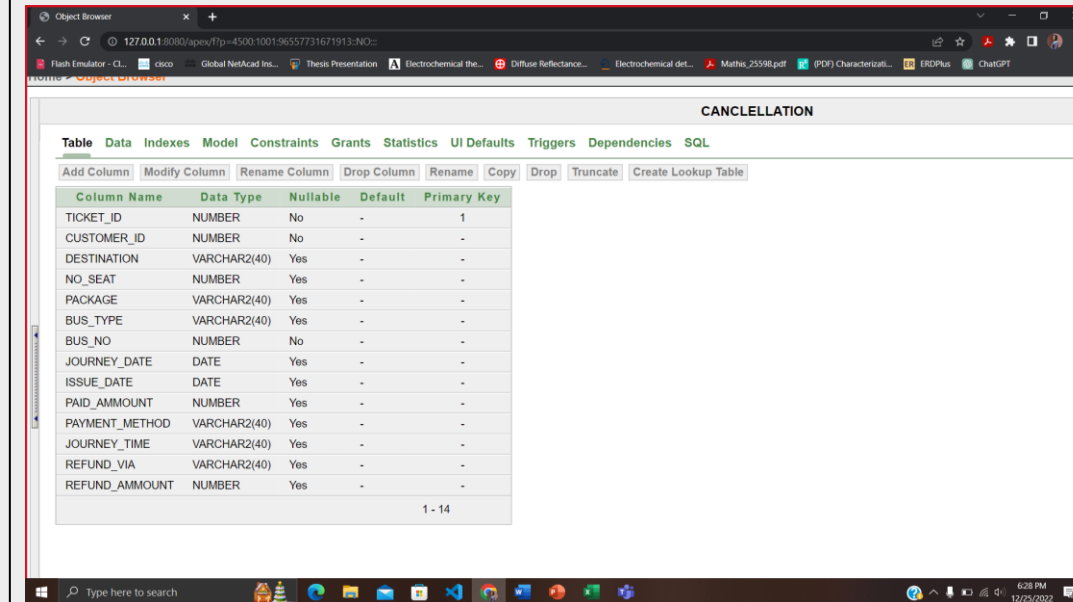
The screenshot shows the Oracle Database Express Edition Object Browser interface. The table 'SOLD_TICKETS' is selected, and its structure is displayed in a table format. The table has 12 columns: TICKET_ID, CUSTOMER_ID, DESTINATION, NO_SEAT, PACKAGE, BUS_TYPE, BUS_NO, JOURNEY_DATE, ISSUE_DATE, PAID_AMMOUNT, PAYMENT_METHOD, and JOURNEY_TIME. The primary key is TICKET_ID. The table is owned by the PROJECT user.

Column Name	Data Type	Nullable	Default	Primary Key
TICKET_ID	NUMBER	No	-	1
CUSTOMER_ID	NUMBER	No	-	-
DESTINATION	VARCHAR2(40)	No	-	-
NO_SEAT	NUMBER	Yes	-	-
PACKAGE	VARCHAR2(40)	Yes	-	-
BUS_TYPE	VARCHAR2(40)	Yes	-	-
BUS_NO	NUMBER	No	-	-
JOURNEY_DATE	DATE	Yes	-	-
ISSUE_DATE	DATE	Yes	-	-
PAID_AMMOUNT	NUMBER	Yes	-	-
PAYMENT_METHOD	VARCHAR2(40)	Yes	-	-
JOURNEY_TIME	VARCHAR2(40)	Yes	-	-

Table creation

CANCELLATION

```
CREATE TABLE "CANCELLATION"
(
    "TICKET_ID" NUMBER NOT NULL ENABLE,
    "CUSTOMER_ID" NUMBER NOT NULL ENABLE,
    "DESTINATION" VARCHAR2(40),
    "NO_SEAT" NUMBER,
    "PACKAGE" VARCHAR2(40),
    "BUS_TYPE" VARCHAR2(40),
    "BUS_NO" NUMBER NOT NULL ENABLE,
    "JOURNEY_DATE" DATE,
    "ISSUE_DATE" DATE,
    "PAID_AMMOUNT" NUMBER,
    "PAYMENT_METHOD" VARCHAR2(40),
    "JOURNEY_TIME" VARCHAR2(40),
    "REFUND_VIA" VARCHAR2(40),
    "REFUND_AMMOUNT" NUMBER,
    CONSTRAINT "CANCELLATION_PK" PRIMARY KEY ("TICKET_ID") ENABLE,
    CONSTRAINT "CANCELLATION_FK" FOREIGN KEY ("CUSTOMER_ID")
    REFERENCES "CUSTOMER" ("C_ID") ENABLE,
    CONSTRAINT "CANCELLATION_FK2" FOREIGN KEY ("PACKAGE")
    REFERENCES "PACKAGE" ("PACKAGE") ENABLE,
    CONSTRAINT "CANCELLATION_FK3" FOREIGN KEY ("BUS_NO")
    REFERENCES "BUS" ("BUS_NO") ENABLE
)
```



The screenshot shows the Oracle SQL Developer interface with the 'CANCELLATION' table selected. The 'Table' tab is active, displaying the table's structure. The table has 14 columns: TICKET_ID, CUSTOMER_ID, DESTINATION, NO_SEAT, PACKAGE, BUS_TYPE, BUS_NO, JOURNEY_DATE, ISSUE_DATE, PAID_AMMOUNT, PAYMENT_METHOD, JOURNEY_TIME, REFUND_VIA, and REFUND_AMMOUNT. The primary key is set to TICKET_ID. The table is located in the schema 'CISCO'.

Column Name	Data Type	Nullable	Default	Primary Key
TICKET_ID	NUMBER	No	-	1
CUSTOMER_ID	NUMBER	No	-	-
DESTINATION	VARCHAR2(40)	Yes	-	-
NO_SEAT	NUMBER	Yes	-	-
PACKAGE	VARCHAR2(40)	Yes	-	-
BUS_TYPE	VARCHAR2(40)	Yes	-	-
BUS_NO	NUMBER	No	-	-
JOURNEY_DATE	DATE	Yes	-	-
ISSUE_DATE	DATE	Yes	-	-
PAID_AMMOUNT	NUMBER	Yes	-	-
PAYMENT_METHOD	VARCHAR2(40)	Yes	-	-
JOURNEY_TIME	VARCHAR2(40)	Yes	-	-
REFUND_VIA	VARCHAR2(40)	Yes	-	-
REFUND_AMMOUNT	NUMBER	Yes	-	-

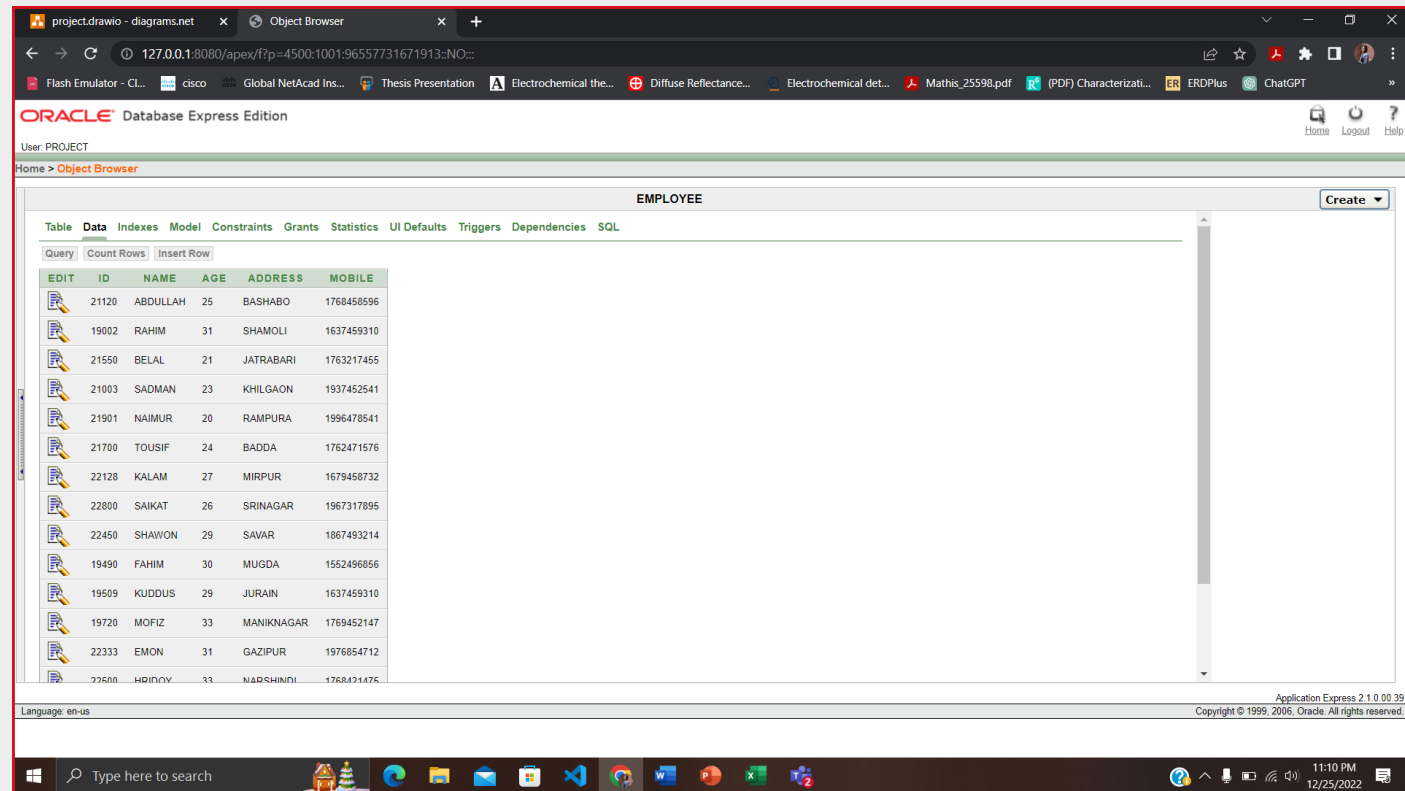
Data Insertion

EMPLPOYEE

INSERT INTO EMPLOYEE VALUES (21120, 'ABDULLAH', 25, 'BASHABO', 01768458596);

▪ ▪ ▪ ▪

▪ ▪ ▪ ▪



The screenshot displays the Oracle Database Express Edition Object Browser interface. The 'EMPLOYEE' table is selected, and the 'Data' tab is active. The table contains 15 rows of data, including the newly inserted record for Abdullah Bashabo. The table structure is as follows:

EDIT	ID	NAME	AGE	ADDRESS	MOBILE
	21120	ABDULLAH	25	BASHABO	1768458596
	19002	RAHIM	31	SHAMOLI	1637459310
	21550	BELAL	21	JATRABARI	1763217455
	21003	SADMAN	23	KHILGAON	1937452541
	21901	NAIMUR	20	RAMPURA	1996478541
	21700	TOUSIF	24	BADDA	1762471576
	22128	KALAM	27	MIRPUR	1679458732
	22800	SAIKAT	26	SRINAGAR	1967317895
	22450	SHAWON	29	SAVAR	1867493214
	19490	FAHIM	30	MUGDA	1552496856
	19509	KUDDUS	29	JURAIN	1637459310
	19720	MOFIZ	33	MANIKNAGAR	1769452147
	22333	EMON	31	GAZIIPUR	1976854712
	22500	HAIRINOV	33	NARSHIMDI	17684721476

At the bottom of the window, the status bar shows 'Language: en-us' and 'Application Express 2.1.0.00.39 Copyright © 1999, 2006, Oracle. All rights reserved.' The Windows taskbar at the very bottom shows the time as 11:10 PM on 12/25/2022.

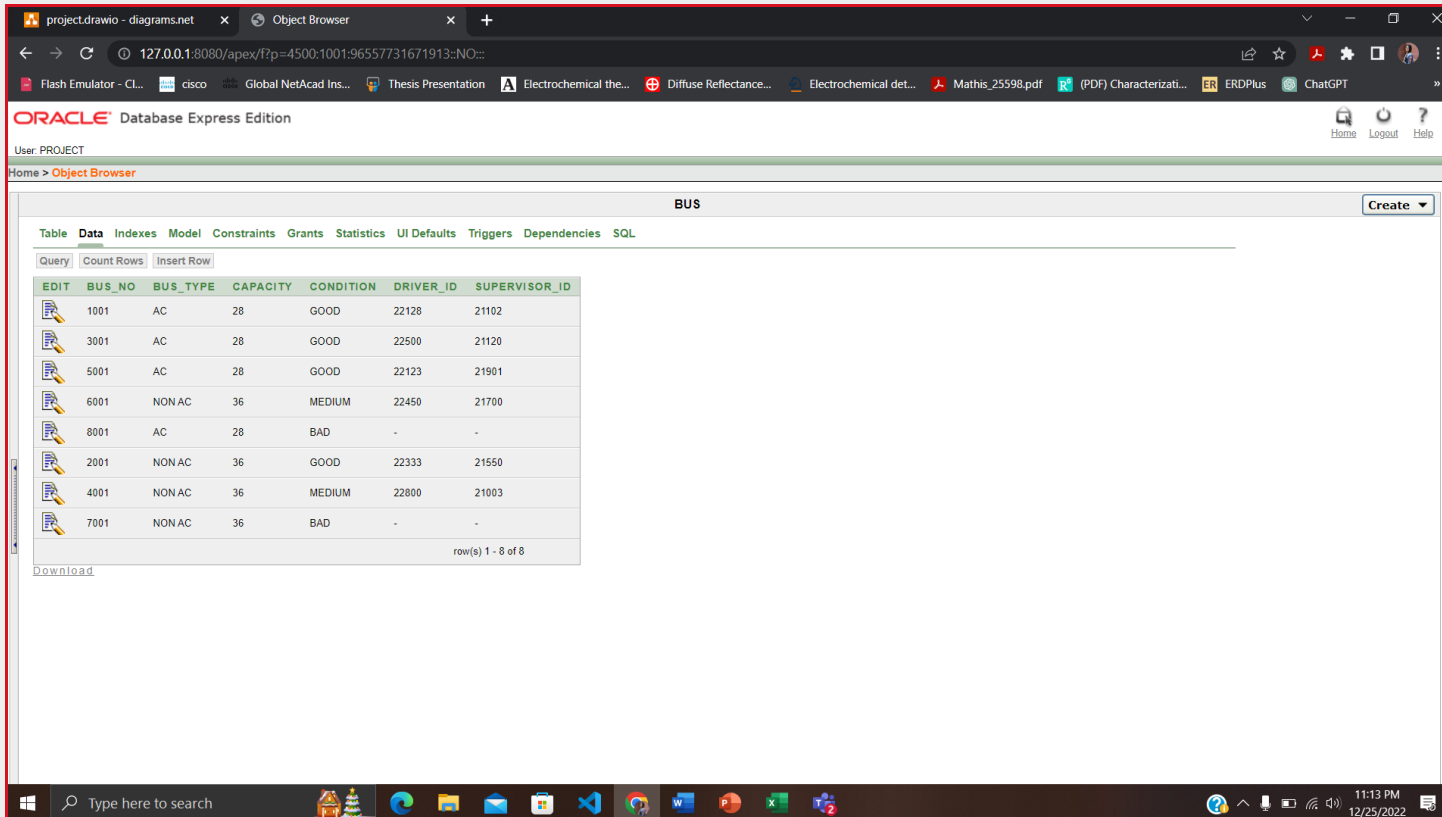
Data Insertion

BUS

```
INSERT INTO BUS VALUES (1001, 'AC', 28, 'GOOD', 22128, 21102,);
```

▪ ▪ ▪ ▪

▪ ▪ ▪ ▪



The screenshot shows the Oracle Database Express Edition Object Browser interface. The 'BUS' table is selected, and the 'Data' tab is active. The table contains 8 rows of data. The columns are BUS_NO, BUS_TYPE, CAPACITY, CONDITION, DRIVER_ID, and SUPERVISOR_ID. The data is as follows:

BUS_NO	BUS_TYPE	CAPACITY	CONDITION	DRIVER_ID	SUPERVISOR_ID
1001	AC	28	GOOD	22128	21102
3001	AC	28	GOOD	22500	21120
5001	AC	28	GOOD	22123	21901
6001	NON AC	36	MEDIUM	22450	21700
8001	AC	28	BAD	-	-
2001	NON AC	36	GOOD	22333	21550
4001	NON AC	36	MEDIUM	22800	21003
7001	NON AC	36	BAD	-	-

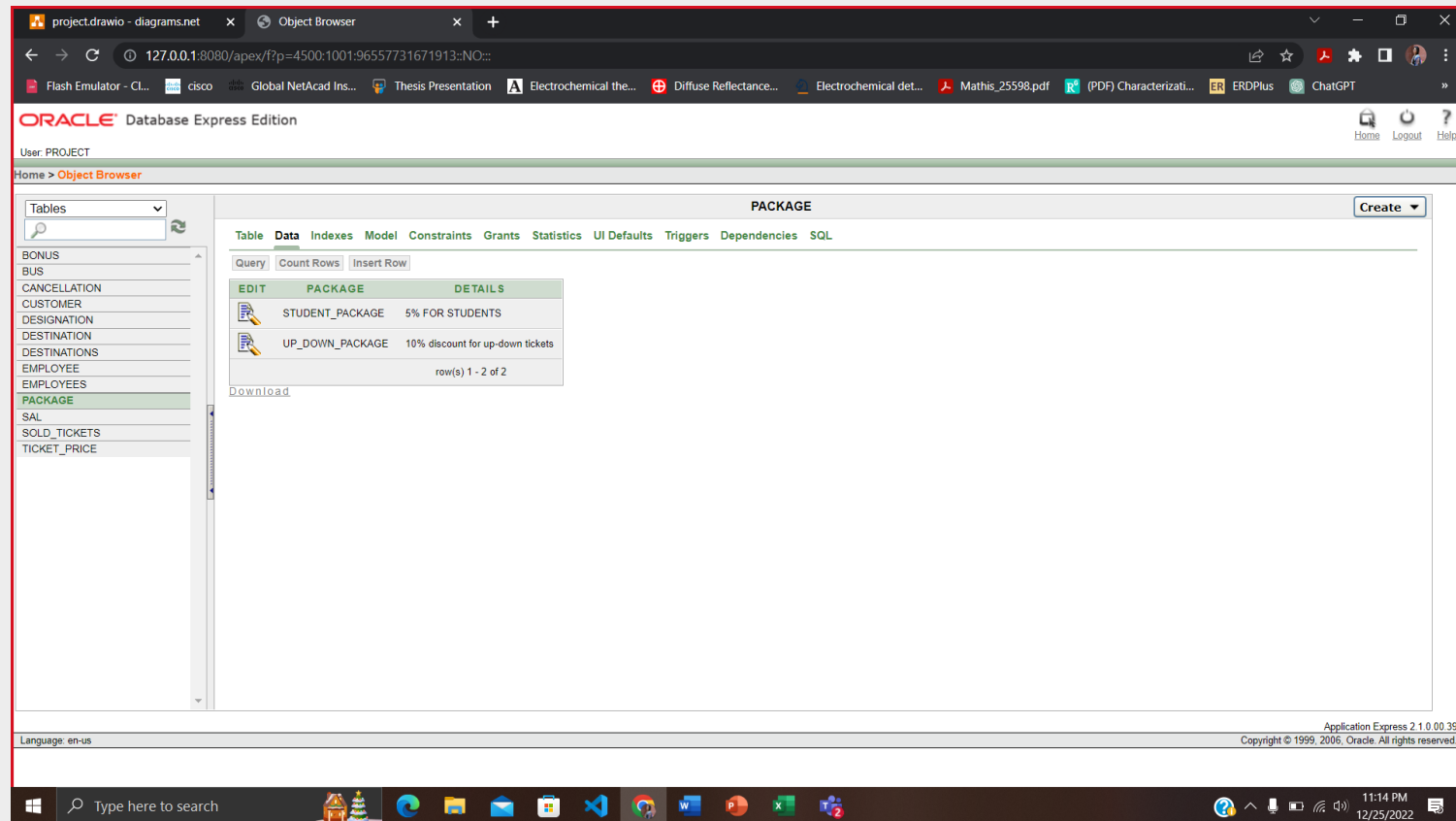
The interface also shows a 'Create' button and a 'Download' link. The status bar at the bottom indicates 'row(s) 1 - 8 of 8'.

Data Insertion

PACKAGE

```
INSERT INTO PACKAGE VALUES ('STUDENT_PACKAGE', '5%FOR STUDENTS');
. . . .
. . . .
```

- ```
INSERT INTO PACKAGE VALUES ('STUDENT_PACKAGE', '5%FOR STUDENTS');
. . . .
. . . .
```



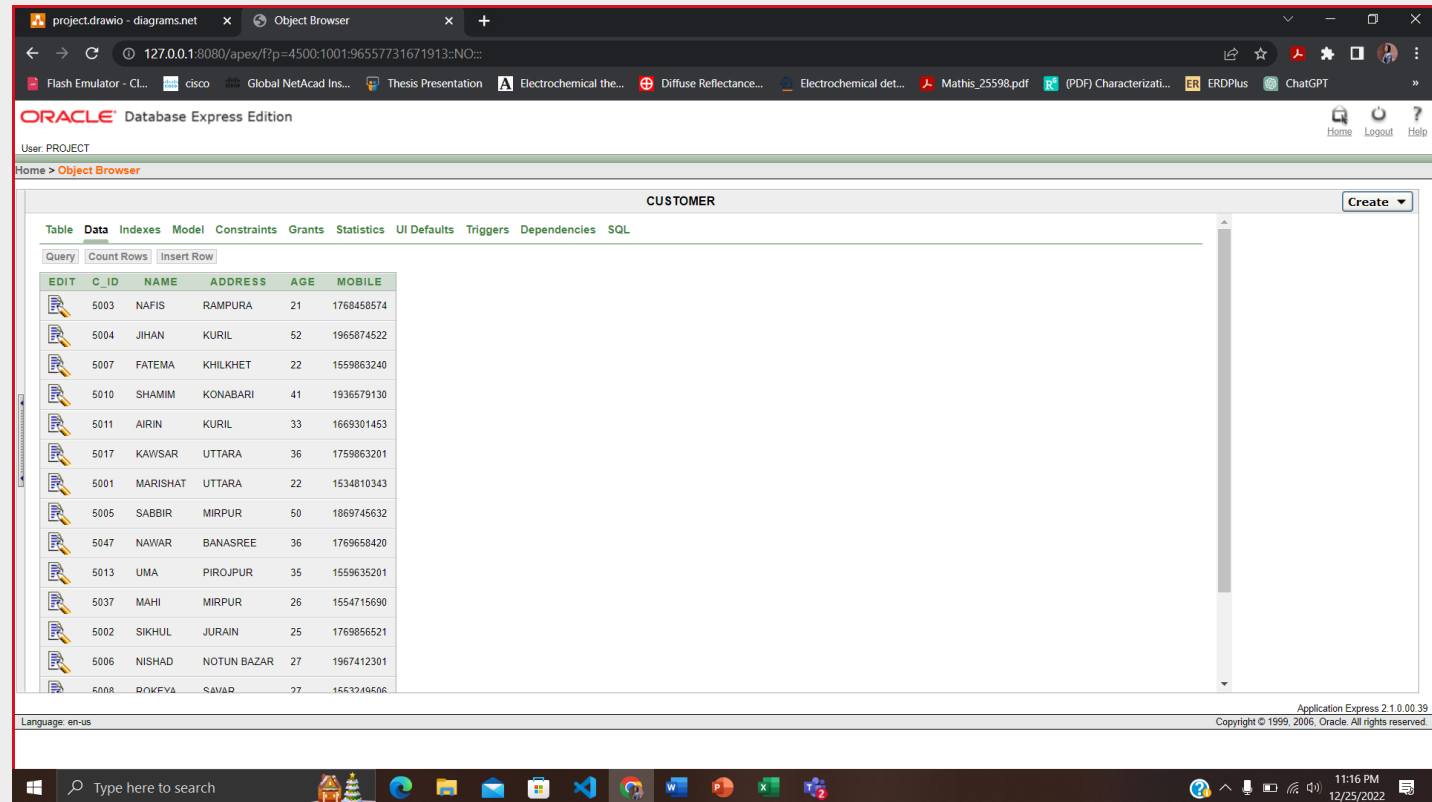
# Data Insertion

# CUSTOMER

INSERT INTO CUSTOMER VALUES (5003, 'NAFIS', 'RAMPURA', 21, 01768458574);

▪ ▪ ▪ ▪

▪ ▪ ▪ ▪



The screenshot displays the Oracle Database Express Edition Object Browser interface. The 'CUSTOMER' table is selected, and the 'Data' tab is active. The table contains 18 rows of data, including the newly inserted row with C\_ID 5003, NAME 'NAFIS', ADDRESS 'RAMPURA', AGE 21, and MOBILE 01768458574. The interface includes a 'Create' button and a 'Language: en-us' footer.

| EDIT | C_ID | NAME     | ADDRESS     | AGE | MOBILE     |
|------|------|----------|-------------|-----|------------|
|      | 5003 | NAFIS    | RAMPURA     | 21  | 1768458574 |
|      | 5004 | JIHAN    | KURIL       | 52  | 1965874522 |
|      | 5007 | FATEMA   | KHILKHET    | 22  | 1559863240 |
|      | 5010 | SHAMIM   | KONABARI    | 41  | 1936579130 |
|      | 5011 | AIRIN    | KURIL       | 33  | 1669301453 |
|      | 5017 | KAWSAR   | UTTARA      | 36  | 1759863201 |
|      | 5001 | MARISHAT | UTTARA      | 22  | 1534810343 |
|      | 5005 | SABBIR   | MIRPUR      | 50  | 1869745632 |
|      | 5047 | NAWAR    | BANASREE    | 36  | 1769650420 |
|      | 5013 | UMA      | PIROJPUR    | 35  | 1559635201 |
|      | 5037 | MAHI     | MIRPUR      | 26  | 1554715690 |
|      | 5002 | SIKHUL   | JURAIN      | 25  | 1769856521 |
|      | 5006 | NISHAD   | NOTUN BAZAR | 27  | 1967412301 |
|      | 5008 | POKEYA   | SAVAR       | 27  | 1663240406 |



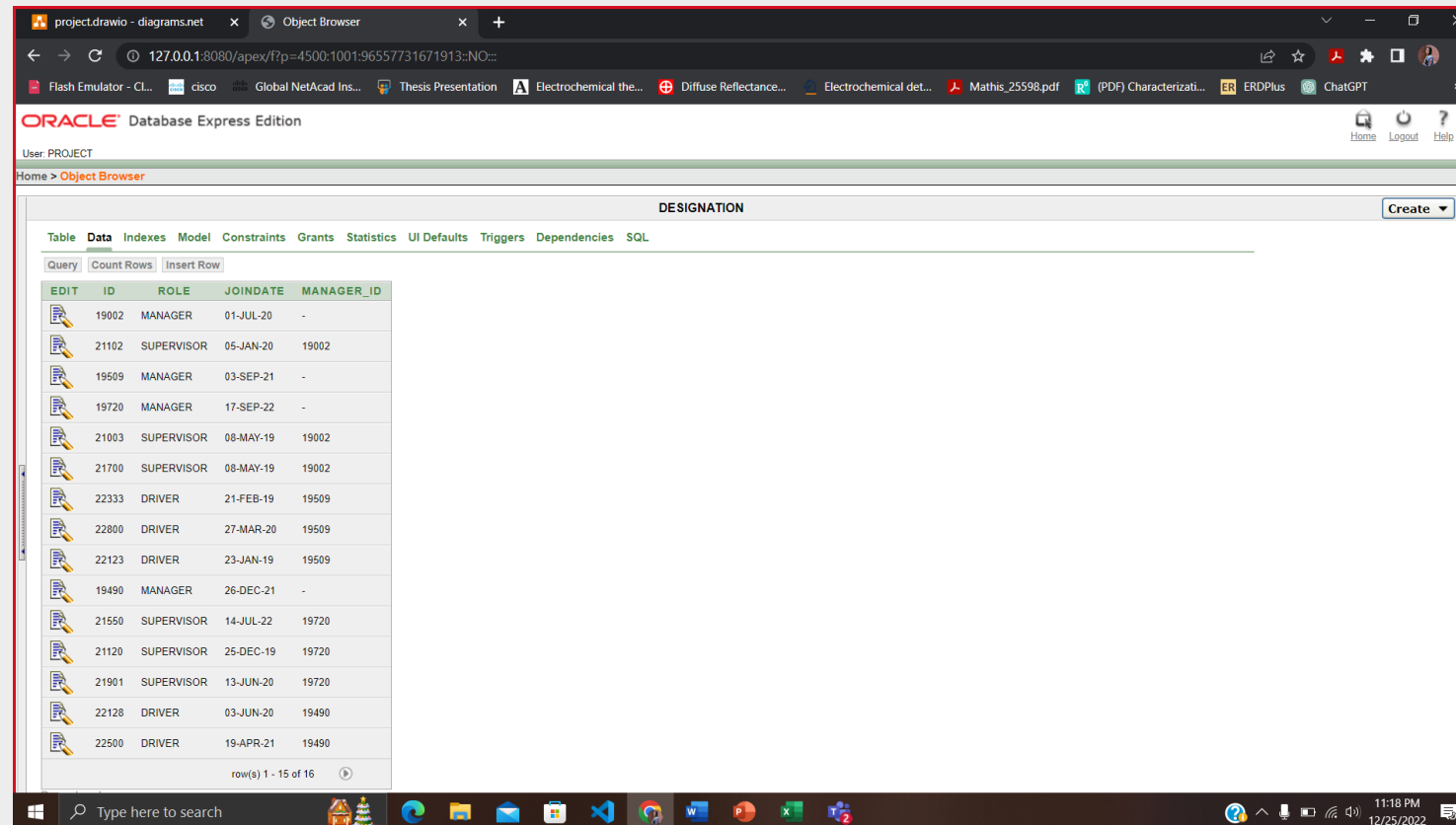
# Data Insertion

# DESIGNATION

INSERT INTO DESIGNATION VALUES (19002, 'MANAGER', 01-JUL-20, ' ');

• • • •

• • • •



The screenshot shows the Oracle Database Express Edition interface. The 'Object Browser' tab is active, displaying the 'DESIGNATION' table. The table has columns: ID, ROLE, JOINDATE, and MANAGER\_ID. The data is as follows:

| EDIT | ID    | ROLE       | JOINDATE  | MANAGER_ID |
|------|-------|------------|-----------|------------|
|      | 19002 | MANAGER    | 01-JUL-20 | -          |
|      | 21102 | SUPERVISOR | 05-JAN-20 | 19002      |
|      | 19509 | MANAGER    | 03-SEP-21 | -          |
|      | 19720 | MANAGER    | 17-SEP-22 | -          |
|      | 21003 | SUPERVISOR | 08-MAY-19 | 19002      |
|      | 21700 | SUPERVISOR | 08-MAY-19 | 19002      |
|      | 22333 | DRIVER     | 21-FEB-19 | 19509      |
|      | 22800 | DRIVER     | 27-MAR-20 | 19509      |
|      | 22123 | DRIVER     | 23-JAN-19 | 19509      |
|      | 19490 | MANAGER    | 26-DEC-21 | -          |
|      | 21550 | SUPERVISOR | 14-JUL-22 | 19720      |
|      | 21120 | SUPERVISOR | 25-DEC-19 | 19720      |
|      | 21901 | SUPERVISOR | 13-JUN-20 | 19720      |
|      | 22128 | DRIVER     | 03-JUN-20 | 19490      |
|      | 22500 | DRIVER     | 19-APR-21 | 19490      |

row(s) 1 - 15 of 16

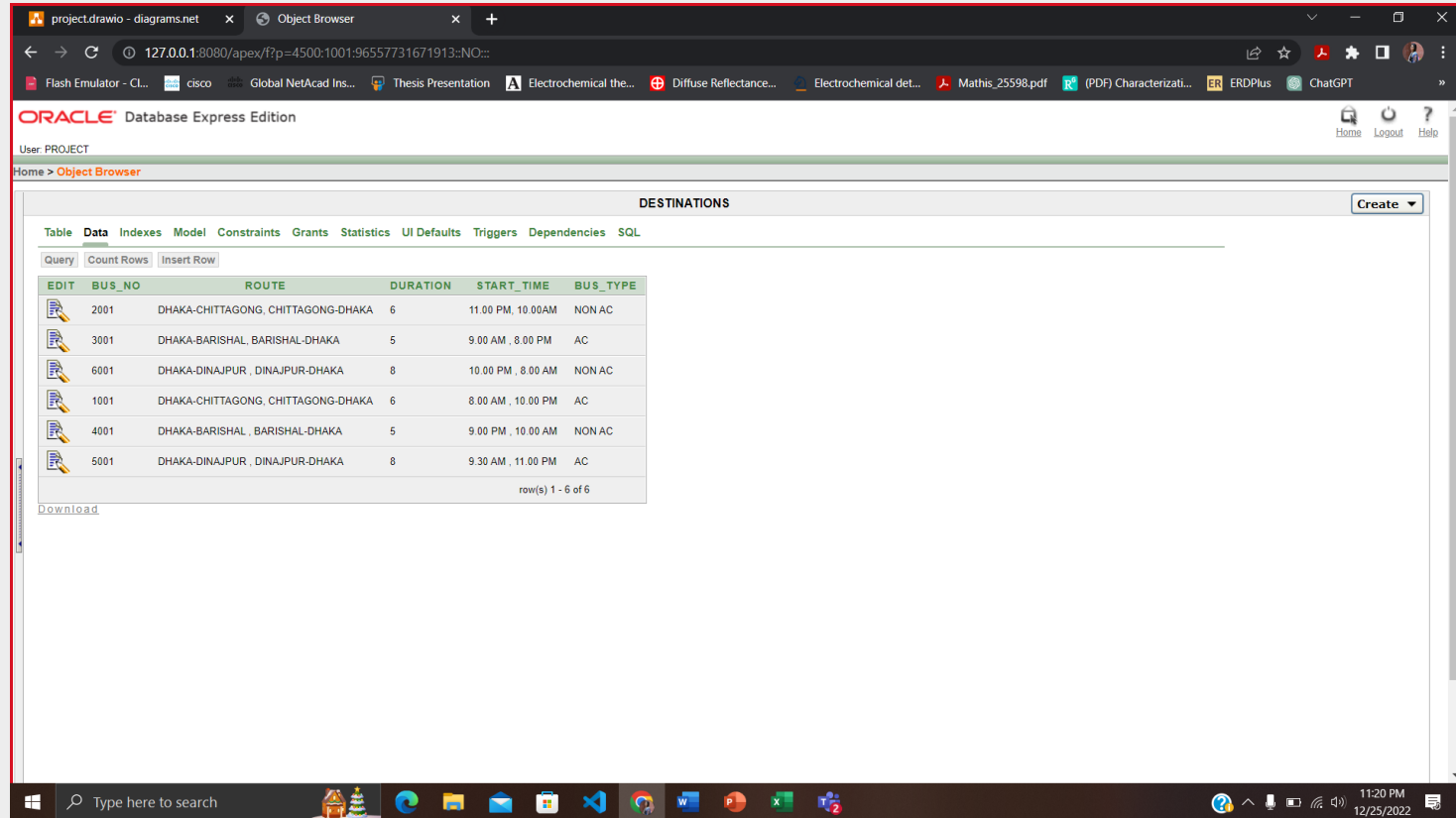
# Data Insertion

# DESTINATION

INSERT INTO DESTINATION VALUES (3001, 'BARISHAL-DHAKA', 5, '8.00 PM', 'AC');

...

...



The screenshot shows the Oracle Database Express Edition interface. The user is logged in as 'PROJECT'. The 'Object Browser' tab is active, displaying the 'DESTINATIONS' table. The table has columns: BUS\_NO, ROUTE, DURATION, START\_TIME, and BUS\_TYPE. The table contains 6 rows of data. The interface includes a 'Create' button and a 'Download' link.

| EDIT | BUS_NO | ROUTE                              | DURATION | START_TIME        | BUS_TYPE |
|------|--------|------------------------------------|----------|-------------------|----------|
|      | 2001   | DHAKA-CHITTAGONG, CHITTAGONG-DHAKA | 6        | 11:00 PM, 10:00AM | NON AC   |
|      | 3001   | DHAKA-BARISHAL, BARISHAL-DHAKA     | 5        | 9:00 AM, 8:00 PM  | AC       |
|      | 6001   | DHAKA-DINAJPUR, DINAJPUR-DHAKA     | 8        | 10:00 PM, 8:00 AM | NON AC   |
|      | 1001   | DHAKA-CHITTAGONG, CHITTAGONG-DHAKA | 6        | 8:00 AM, 10:00 PM | AC       |
|      | 4001   | DHAKA-BARISHAL, BARISHAL-DHAKA     | 5        | 9:00 PM, 10:00 AM | NON AC   |
|      | 5001   | DHAKA-DINAJPUR, DINAJPUR-DHAKA     | 8        | 9:30 AM, 11:00 PM | AC       |

row(s) 1 - 6 of 6

Download

# Data Insertion

# SAL

INSERT INTO SAL VALUES (19002, 'MANAGER', 224000);

■ ■ ■ ■

■ ■ ■ ■

The screenshot displays the Oracle Database Express Edition Object Browser interface. The browser window shows the 'SAL' table under the 'PROJECT' user. The table has columns: ID, ROLE, and SAL. The data is as follows:

| ID    | ROLE       | SAL    |
|-------|------------|--------|
| 19002 | MANAGER    | 224000 |
| 21102 | SUPERVISOR | 210000 |
| 19509 | MANAGER    | 264000 |
| 21550 | SUPERVISOR | 213000 |
| 21120 | SUPERVISOR | 200200 |
| 21003 | SUPERVISOR | 220600 |
| 21901 | SUPERVISOR | 230000 |
| 22500 | DRIVER     | 365000 |
| 22800 | DRIVER     | 326067 |
| 22450 | DRIVER     | 325000 |
| 19490 | MANAGER    | 244000 |
| 19720 | MANAGER    | 274000 |
| 21700 | SUPERVISOR | 200000 |
| 22333 | DRIVER     | 325600 |

The interface includes a 'Create' button and a 'Language: en-us' footer. The bottom of the screen shows a Windows taskbar with various application icons and a system clock indicating 11:21 PM on 12/25/2022.

INSERT INTO BONUS VALUES (21003, '3%');

...

...

The screenshot shows the Oracle Database Express Edition Object Browser interface. The browser is displaying the 'BONUS' table under the 'PROJECT' user. The table has two columns: 'ID' and 'BONUS'. The data is as follows:

| ID    | BONUS |
|-------|-------|
| 21003 | 3%    |
| 19002 | 5%    |
| 21550 | 3%    |
| 22123 | 5%    |
| 22500 | 3%    |
| 22128 | 5%    |
| 19720 | 5%    |

The interface includes a 'Create' button and a 'Download' link. The bottom of the window shows the Windows taskbar with the time 11:22 PM on 12/25/2022.

# Data Insertion

# TICKET PRICE

```
INSERT INTO TICKET_PRICE VALUES ('DHAKA-BARISHAL', 1100, 700);
```

■ ■ ■ ■

■ ■ ■ ■

The screenshot displays the Oracle Database Express Edition interface. The browser address bar shows the URL `127.0.0.1:8080/apex/f?p=4500:1001:96557731671913::NO::`. The page title is "ORACLE Database Express Edition" and the user is "PROJECT". The navigation bar includes "Home", "Logout", and "Help". The main content area shows the "TICKET\_PRICE" table with tabs for "Table", "Data", "Indexes", "Model", "Constraints", "Grants", "Statistics", "UI Defaults", "Triggers", "Dependencies", and "SQL". The "Data" tab is selected, showing a table with columns "EDIT", "DESTINATION", "AC", and "NON\_AC". The table contains 6 rows of data, with the first row being "DHAKA-BARISHAL" with values 1100 and 700. The status bar at the bottom indicates "Application Express 2.1.0.00.39" and "Copyright © 1999, 2006, Oracle. All rights reserved."

| EDIT | DESTINATION      | AC   | NON_AC |
|------|------------------|------|--------|
|      | DHAKA-BARISHAL   | 1100 | 700    |
|      | DHAKA-CHITTAGONG | 1200 | 600    |
|      | CHITTAGONG-DHAKA | 1200 | 600    |
|      | DHAKA-DINAJPUR   | 1500 | 800    |
|      | BARISHAL-DHAKA   | 1100 | 700    |
|      | DINAJPUR-DHAKA   | 1500 | 800    |

row(s) 1 - 6 of 6

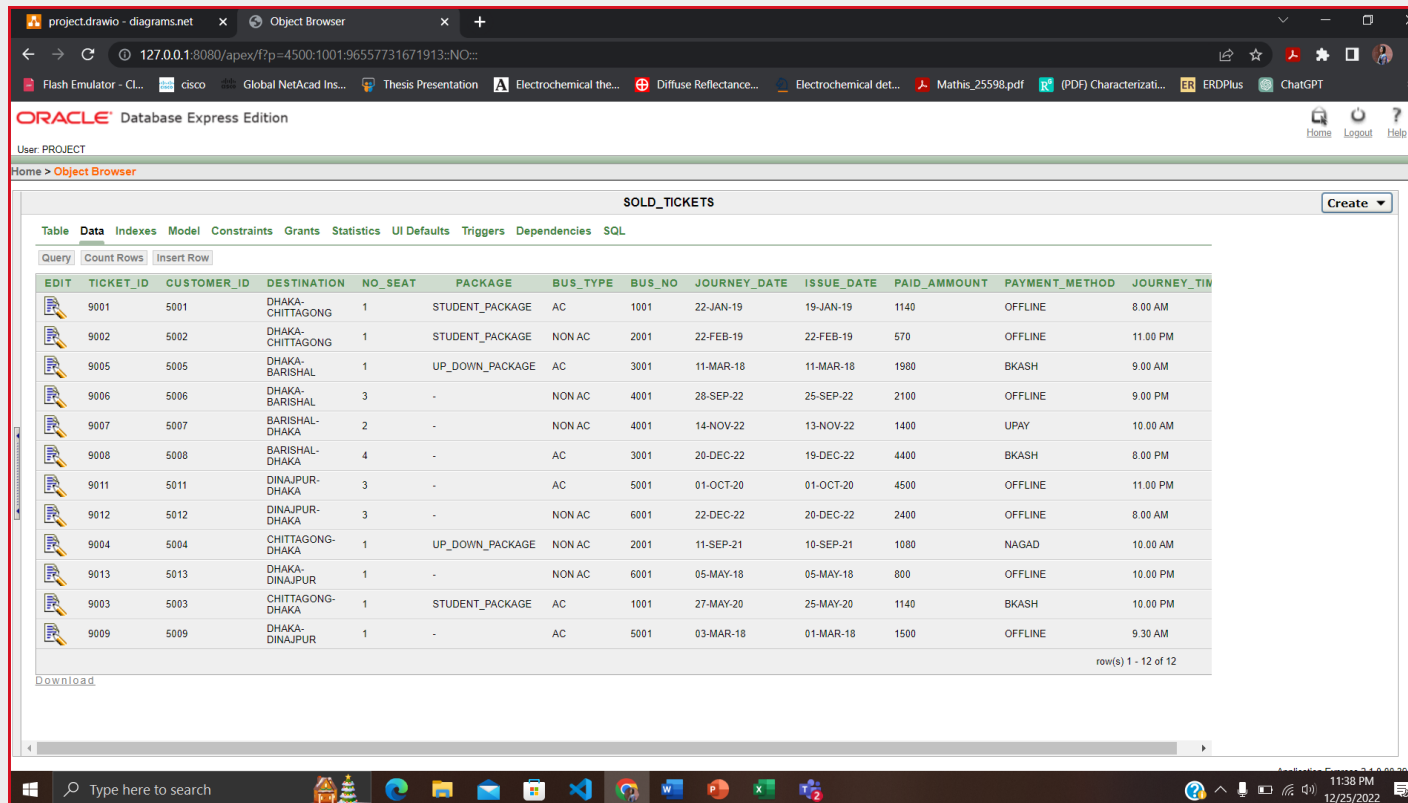
Download

# Data Insertion

# SOLD TICKET

INSERT INTO SOLD\_TICKETS VALUES (9001, 5001, 'DHAKA-CHITTAGONG', 1, 'STUDEN\_PACKAGE', 'AC', 1001, 22-JAN-19, 19-JAN-19, 1140, 'OFFLINE', '8.00 AM');

...



The screenshot displays the Oracle Database Express Edition interface. The 'SOLD\_TICKETS' table is selected, and the 'Data' tab is active. The table contains 12 rows of data. The columns are: TICKET\_ID, CUSTOMER\_ID, DESTINATION, NO\_SEAT, PACKAGE, BUS\_TYPE, BUS\_NO, JOURNEY\_DATE, ISSUE\_DATE, PAID\_AMMOUNT, PAYMENT\_METHOD, and JOURNEY\_TIM. The data is as follows:

| TICKET_ID | CUSTOMER_ID | DESTINATION      | NO_SEAT | PACKAGE         | BUS_TYPE | BUS_NO | JOURNEY_DATE | ISSUE_DATE | PAID_AMMOUNT | PAYMENT_METHOD | JOURNEY_TIM |
|-----------|-------------|------------------|---------|-----------------|----------|--------|--------------|------------|--------------|----------------|-------------|
| 9001      | 5001        | DHAKA-CHITTAGONG | 1       | STUDENT_PACKAGE | AC       | 1001   | 22-JAN-19    | 19-JAN-19  | 1140         | OFFLINE        | 8.00 AM     |
| 9002      | 5002        | DHAKA-CHITTAGONG | 1       | STUDENT_PACKAGE | NON AC   | 2001   | 22-FEB-19    | 22-FEB-19  | 570          | OFFLINE        | 11.00 PM    |
| 9005      | 5005        | DHAKA-BARISHAL   | 1       | UP_DOWN_PACKAGE | AC       | 3001   | 11-MAR-18    | 11-MAR-18  | 1980         | BKASH          | 9.00 AM     |
| 9006      | 5006        | DHAKA-BARISHAL   | 3       | -               | NON AC   | 4001   | 28-SEP-22    | 25-SEP-22  | 2100         | OFFLINE        | 9.00 PM     |
| 9007      | 5007        | BARISHAL-DHAKA   | 2       | -               | NON AC   | 4001   | 14-NOV-22    | 13-NOV-22  | 1400         | UPAY           | 10.00 AM    |
| 9008      | 5008        | BARISHAL-DHAKA   | 4       | -               | AC       | 3001   | 20-DEC-22    | 19-DEC-22  | 4400         | BKASH          | 8.00 PM     |
| 9011      | 5011        | DINAJPUR-DHAKA   | 3       | -               | AC       | 5001   | 01-OCT-20    | 01-OCT-20  | 4500         | OFFLINE        | 11.00 PM    |
| 9012      | 5012        | DINAJPUR-DHAKA   | 3       | -               | NON AC   | 6001   | 22-DEC-22    | 20-DEC-22  | 2400         | OFFLINE        | 8.00 AM     |
| 9004      | 5004        | CHITTAGONG-DHAKA | 1       | UP_DOWN_PACKAGE | NON AC   | 2001   | 11-SEP-21    | 10-SEP-21  | 1080         | NAGAD          | 10.00 AM    |
| 9013      | 5013        | DHAKA-DINAJPUR   | 1       | -               | NON AC   | 6001   | 05-MAY-18    | 05-MAY-18  | 800          | OFFLINE        | 10.00 PM    |
| 9003      | 5003        | CHITTAGONG-DHAKA | 1       | STUDENT_PACKAGE | AC       | 1001   | 27-MAY-20    | 25-MAY-20  | 1140         | BKASH          | 10.00 PM    |
| 9009      | 5009        | DHAKA-DINAJPUR   | 1       | -               | AC       | 5001   | 03-MAR-18    | 01-MAR-18  | 1500         | OFFLINE        | 9.30 AM     |

# Data Insertion

# CANCELLATION

INSERT INTO CANCELLATION VALUES (9027, 5027, 'DHAKA-CHITTAGONG', 1, ' ', 'AC', 1001, 26-JUL-20, 25-JUL-20, 1200, 'BKASH', 8.00 AM,'BKASH',1080);

- ▪ ▪ ▪
- ▪ ▪ ▪

The screenshot shows the Oracle Database Express Edition interface. The 'Object Browser' tab is active, displaying the 'CANCELLATION' table. The table has 12 columns: DESTINATION, NO\_SEAT, PACKAGE, BUS\_TYPE, BUS\_NO, JOURNEY\_DATE, ISSUE\_DATE, PAID\_AMMOUNT, PAYMENT\_METHOD, JOURNEY\_TIME, REFUND\_VIA, and REFUND\_AMMOUNT. The table contains 5 rows of data. The bottom of the window shows the Windows taskbar with the date 12/25/2022 and time 11:41 PM.

| DESTINATION      | NO_SEAT | PACKAGE         | BUS_TYPE | BUS_NO | JOURNEY_DATE | ISSUE_DATE | PAID_AMMOUNT | PAYMENT_METHOD | JOURNEY_TIME | REFUND_VIA | REFUND_AMMOUNT |
|------------------|---------|-----------------|----------|--------|--------------|------------|--------------|----------------|--------------|------------|----------------|
| DHAKA-CHITTAGONG | 1       | -               | AC       | 1001   | 26-JUL-20    | 25-JUL-20  | 1200         | BKASH          | 8.00 AM      | BKASH      | 1080           |
| BARISHAL-DHAKA   | 2       | -               | AC       | 3001   | 15-MAY-18    | 13-MAY-18  | 2200         | OFFLINE        | 8.00 PM      | OFFLINE    | 1980           |
| DHAKA-BARISHAL   | 1       | UP_DOWN_PACKAGE | AC       | 3001   | 13-MAR-18    | 10-MAR-18  | 1980         | NAGAD          | 9.00 AM      | NAGAD      | 1782           |
| DHAKA-DINAJPUR   | 1       | -               | NON AC   | 6001   | 05-MAY-18    | 05-MAY-18  | 800          | NAGAD          | 10.00 PM     | NAGAD      | 720            |
| CHITTAGONG-DHAKA | 2       | -               | NON AC   | 2001   | 20-NOV-19    | 19-NOV-19  | 1200         | UPAY           | 10.00 AM     | UPAY       | 1080           |

Write a query to find those employee's name whose age are under 25?

```
SELECT NAME FROM
EMPLOYEE WHERE AGE<25
```

The screenshot shows the Oracle Database Express Edition interface. The SQL Commands window contains the query: `SELECT NAME FROM EMPLOYEE WHERE AGE<25`. The Results tab is active, displaying a table with 5 rows of employee names: BELAL, SADMAN, NAIMUR, TOUSIF, and KARIM. The status bar indicates "5 rows returned in 0.00 seconds".

| NAME   |
|--------|
| BELAL  |
| SADMAN |
| NAIMUR |
| TOUSIF |
| KARIM  |

5 rows returned in 0.00 seconds [CSV Export](#)

Language: en-us

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Slide 14 of 15 English (United States) Accessibility: Investigate



Write a query to find the Maximum and Minimum number of seats were sold?

```
SELECT
MAX(NO_SEAT),MIN(NO_SEAT)
FROM SOLD_TICKETS
```

The screenshot shows the Oracle Database Express Edition interface. The SQL command window contains the following query:

```
SELECT MAX(NO_SEAT),MIN(NO_SEAT) FROM SOLD_TICKETS |
```

The query has been executed, and the results are displayed in a table:

| MAX(NO_SEAT) | MIN(NO_SEAT) |
|--------------|--------------|
| 4            | 1            |

Below the table, it states: "1 rows returned in 0.00 seconds" and provides a "CSV Export" link.

The interface also shows the user "PROJECT" and the "SQL Commands" tab. The bottom status bar indicates "Application Express 2.1.0.00.39" and "Copyright © 1999, 2006, Oracle. All rights reserved."

Write a query to find the oldest customer?

```
SELECT * FROM CUSTOMER
WHERE AGE=(SELECT
MAX(AGE) FROM CUSTOMER)
```

The screenshot shows the Oracle Database Express Edition interface. The SQL command window contains the query: `SELECT * FROM CUSTOMER WHERE AGE=(SELECT MAX(AGE) FROM CUSTOMER)`. The results window displays a single row of data for the oldest customer.

| C_ID | NAME  | ADDRESS | AGE | MOBILE     |
|------|-------|---------|-----|------------|
| 5004 | JIHAN | KURIL   | 52  | 1965874522 |

1 rows returned in 0.02 seconds

Application Express 2.1.0.00.39  
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Write a query to find the information of those employee's whose salary is more than average salary?

```
SELECT ID,ROLE,SAL FROM SAL WHERE
SAL>(SELECT AVG(SAL) FROM SAL)
```

The screenshot shows the Oracle Database Express Edition interface. The SQL Commands window contains the following query:

```
SELECT ID,ROLE,SAL FROM SAL WHERE SAL>(SELECT AVG(SAL) FROM SAL)
```

The query has been executed, and the results are displayed in a table with 6 rows. The columns are ID, ROLE, and SAL. The results are as follows:

| ID    | ROLE   | SAL    |
|-------|--------|--------|
| 22500 | DRIVER | 325000 |
| 22800 | DRIVER | 325000 |
| 22450 | DRIVER | 325000 |
| 22333 | DRIVER | 325000 |
| 22128 | DRIVER | 325000 |
| 22123 | DRIVER | 325000 |

6 rows returned in 0.00 seconds

Language: en-us

Application Express 2.1.0.00.39  
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Write a query to find the employee who gets role wise lowest salary ?

```
SELECT ID,ROLE FROM SAL WHERE
SAL IN(SELECT MIN(SAL) FROM SAL
GROUP BY ROLE)
```

The screenshot displays the Oracle Database Express Edition interface. The SQL command window contains the following query:

```
SELECT ID,ROLE FROM SAL WHERE SAL IN(SELECT MIN(SAL) FROM SAL GROUP BY ROLE)
```

The query has been executed, and the results are shown in a table with 3 rows returned in 0.00 seconds. The table has two columns: ID and ROLE.

| ID    | ROLE       |
|-------|------------|
| 19002 | MANAGER    |
| 21700 | SUPERVISOR |
| 22128 | DRIVER     |

The interface also shows the 'Results' tab selected, and the 'Run' button was used to execute the query. The bottom status bar indicates 'Application Express 2.1.0.0.39' and 'Copyright © 1999, 2006, Oracle. All rights reserved.'

Write a query to find those customer's id who canceled their tickets?

```
SELECT CUSTOMER_ID FROM
CANCELLATION WHERE TICKET_ID
IN(SELECT TICKET_ID FROM CANCELLATION)
```

The screenshot shows the Oracle Database Express Edition SQL Commands window. The query entered is: `SELECT CUSTOMER_ID FROM CANCELLATION WHERE TICKET_ID IN(SELECT TICKET_ID FROM CANCELLATION)`. The results are displayed in a table with the column `CUSTOMER_ID` and 5 rows of data: 5027, 5017, 5037, 5010, and 5047. The status bar indicates "5 rows returned in 0.00 seconds".

| CUSTOMER_ID |
|-------------|
| 5027        |
| 5017        |
| 5037        |
| 5010        |
| 5047        |

**Write a query to find all Bus no , their Route and their Driver id?**

```
SELECT BUS.BUS_NO, DESTINATION.ROUTE, BUS.DRIVER_ID FROM
BUS,DESTINATION WHERE BUS.BUS_NO=DESTINATION.BUS_NO
```

The screenshot shows the Oracle Database Express Edition interface. The SQL Commands window contains the following query:

```
SELECT BUS.BUS_NO,DESTINATION.ROUTE,BUS.DRIVER_ID FROM BUS,DESTINATION WHERE BUS.BUS_NO=DESTINATION.BUS_NO
```

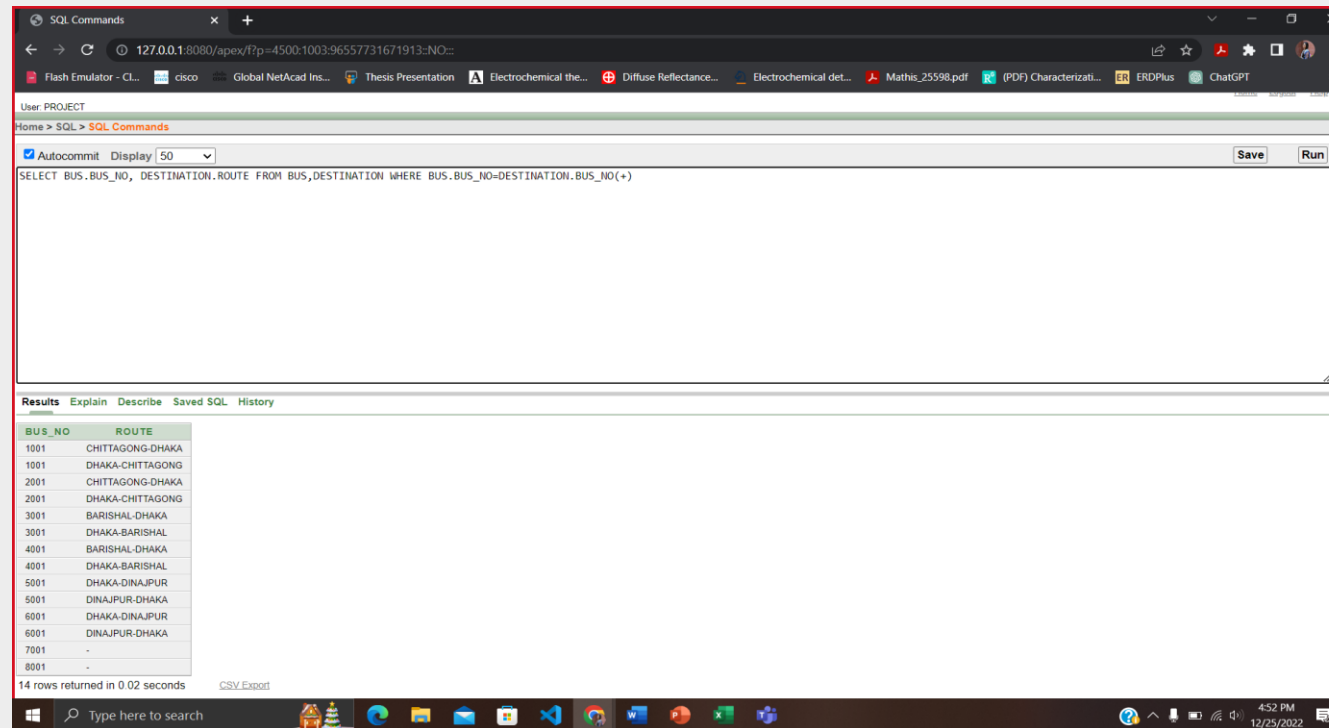
The Results tab displays the following data:

| BUS_NO | ROUTE            | DRIVER_ID |
|--------|------------------|-----------|
| 1001   | CHITTAGONG-DHAKA | 22128     |
| 1001   | DHAKA-CHITTAGONG | 22128     |
| 3001   | BARISHAL-DHAKA   | 22500     |
| 3001   | DHAKA-BARISHAL   | 22500     |
| 5001   | DHAKA-DINAJPUR   | 22123     |
| 5001   | DINAJPUR-DHAKA   | 22123     |
| 6001   | DHAKA-DINAJPUR   | 22450     |
| 6001   | DINAJPUR-DHAKA   | 22450     |
| 2001   | CHITTAGONG-DHAKA | 22333     |
| 2001   | DHAKA-CHITTAGONG | 22333     |

More than 10 rows available. Increase rows selector to view more rows.  
10 rows returned in 0.00 seconds

**Write a query to find all bus no and their route including both available and unavailable buses?**

```
SELECT BUS.BUS_NO, DESTINATION.ROUTE FROM BUS,DESTINATION WHERE
BUS.BUS_NO=DESTINATION.BUS_NO(+)
```



The screenshot shows a web-based SQL interface. The query entered is: `SELECT BUS.BUS_NO, DESTINATION.ROUTE FROM BUS,DESTINATION WHERE BUS.BUS_NO=DESTINATION.BUS_NO(+)`. The results are displayed in a table with 14 rows. The first 13 rows show bus numbers 1001 through 8001 with their respective routes. The last row (8001) shows a dash '-' for the route.

| BUS_NO | ROUTE            |
|--------|------------------|
| 1001   | CHITTAGONG-DHAKA |
| 1001   | DHAKA-CHITTAGONG |
| 2001   | CHITTAGONG-DHAKA |
| 2001   | DHAKA-CHITTAGONG |
| 3001   | BARISHAL-DHAKA   |
| 3001   | DHAKA-BARISHAL   |
| 4001   | BARISHAL-DHAKA   |
| 4001   | DHAKA-BARISHAL   |
| 5001   | DHAKA-DINAJPUR   |
| 5001   | DINAJPUR-DHAKA   |
| 6001   | DHAKA-DINAJPUR   |
| 6001   | DINAJPUR-DHAKA   |
| 7001   | -                |
| 8001   | -                |

14 rows returned in 0.02 seconds

Write a query to find the manager id of all employee's?

```
SELECT WORKER.ID || ' WORKS FOR ' || MANAGER.ID FROM DESIGNATION
WORKER, DESIGNATION MANAGER WHERE WORKER.MANAGER_ID = MANAGER.ID
```

The screenshot shows the Oracle Database Express Edition interface. The SQL command window contains the following query:

```
SELECT WORKER.ID || ' WORKS FOR ' || MANAGER.ID FROM DESIGNATION
WORKER, DESIGNATION MANAGER WHERE WORKER.MANAGER_ID = MANAGER.ID
```

The query has been executed, and the results are displayed in a table with the following columns: **WORKER.ID||'WORKSFOR'||MANAGER.ID**. The results show 12 rows of data, each representing an employee and their manager.

| WORKER.ID  'WORKSFOR'  MANAGER.ID |
|-----------------------------------|
| 21102 WORKS FOR 19002             |
| 21003 WORKS FOR 19002             |
| 21700 WORKS FOR 19002             |
| 22333 WORKS FOR 19509             |
| 22800 WORKS FOR 19509             |
| 22123 WORKS FOR 19509             |
| 21550 WORKS FOR 19720             |
| 21120 WORKS FOR 19720             |
| 21901 WORKS FOR 19720             |
| 22128 WORKS FOR 19490             |
| 22500 WORKS FOR 19490             |
| 22450 WORKS FOR 19490             |

12 rows returned in 0.00 seconds



TO CREATE VIEW FOR BUS TABLE :

CREATE OR REPLACE VIEW BUSVIEW AS SELECT BUS\_NO,BUS\_TYPE,CAPACITY FROM BUS WITH READ ONLY

TO SEE THE VIEW :

SELECT \* FROM BUSVIEW

The screenshot shows the Oracle Database Express Edition interface. The SQL Commands window is open, displaying the query `SELECT * FROM BUSVIEW`. The query has been executed, and the results are shown in a table with 8 rows. The table has columns `BUS_NO`, `BUS_TYPE`, and `CAPACITY`. The results are as follows:

| BUS_NO | BUS_TYPE | CAPACITY |
|--------|----------|----------|
| 1001   | AC       | 28       |
| 3001   | AC       | 28       |
| 5001   | AC       | 28       |
| 6001   | NON AC   | 36       |
| 8001   | AC       | 28       |
| 2001   | NON AC   | 36       |
| 4001   | NON AC   | 36       |
| 7001   | NON AC   | 36       |

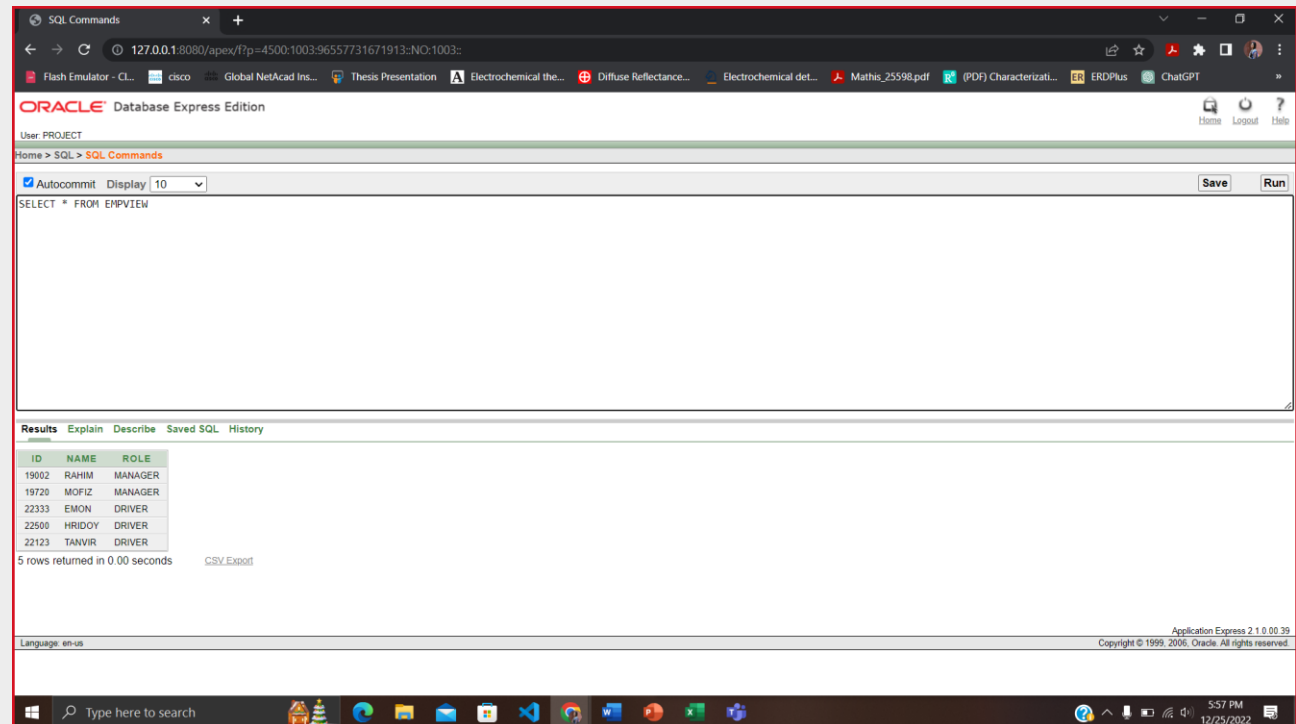
8 rows returned in 0.01 seconds

TO CREATE VIEW FOR EMPLOYEE TABLE :

CREATE OR REPLACE VIEW EMPVIEW AS SELECT  
EMPLOYEE.ID,EMPLOYEE.NAME,DESIGNATION.ROLE FROM EMPLOYEE,DESIGNATION  
WHERE EMPLOYEE.ID=DESIGNATION.ID AND EMPLOYEE.AGE>30 WITH READ ONLY

TO SEE THE VIEW :

SELECT \* FROM EMPVIEW



The screenshot shows the Oracle Database Express Edition interface. The SQL Commands window is active, displaying the query: `SELECT * FROM EMPVIEW`. The query has been executed successfully, and the results are displayed in a table below. The table has three columns: ID, NAME, and ROLE. The results show 5 rows of data.

| ID    | NAME   | ROLE    |
|-------|--------|---------|
| 19002 | RAHIM  | MANAGER |
| 19720 | MOFIZ  | MANAGER |
| 22333 | EMON   | DRIVER  |
| 22600 | HRIDOY | DRIVER  |
| 22123 | TANVIR | DRIVER  |

5 rows returned in 0.00 seconds



## Conclusion & Future works :

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Finally, the Bus Management Database Project has established a comprehensive database system for managing and tracking the business of a bus transportation company. This project has the potential to increase bus transportation operations and management fluently in the future by including more features such as ticket sales and fare administration, adopting real-time tracking, and creating machine learning algorithms.

## Thank you. . .